Online Data Supplement

Clinical utility of diagnostic biomarkers in malignant pleural mesothelioma:
A systematic review and meta-analysis.

Eline Schillebeeckx, MSc, Jan P. van Meerbeeck, MD, Kevin Lamote, PhD

Interesting markers for which a meta-analysis was not possible are described in the supplementary text below.

SERUM

a. Lipids

Despite their role in membrane structures, energy storage, signal transduction and cancer development, lipids have not yet been extensively investigated as MPM biomarkers (1). One study revealed a promising result for several lipids by using liquid chromatography high-resolution mass spectrometry (2). An AUC_{ROC} of 0.991, 0.988 and 0.993 was obtained for respectively L379.28/L372.31, L372.31 and L372.31/L782.57. Unfortunately, only the mass-to-charge ratio of each lipid is given and the correlated compounds were not identified.

b. Circulating endothelial cells (CEC)/Circulating tumour cells (CTC)

CECs are endothelial cells that are released from the vascular wall and circulate in the peripheral blood, while CTCs are tumour cells that have shed from a primary tumour. Higher levels of both have been found in MPM patients compared to controls (3, 4). While patients under suspicion but excluded from having MPM were recruited for the CTC research, only people with benign pleural disease were included as control group when studying CEC's. With a cut-off of 50CEC's/4.0ml, 49.9% sensitivity and 76.7% specificity were obtained. For CTC's, a higher specificity (90.6%), but lower sensitivity (32.7%) were found at a cut-off level of one CTC. As seen with many biomarkers, CTC's were more prominent in the epithelioid subtype (3).

PLASMA

a. miRNA

In total, 5 studies focused on miRNA's in plasma. All miRNA's (miR-132-3p, miR-126, miR-17, miR-486, miR16, and miR-103) were found to be downregulated, except for miR625-3p (5). For this miRNA, a comparison between MPM and HC resulted in an AUC_{ROC} of 0.82 (5). The best diagnostic accuracy (AUC_{ROC}=0.95) was obtained by miR-126 when discriminating MPM from asbestosis and BRD patients (6). Further research is necessary to confirm this result.

PLEURAL EFFUSION

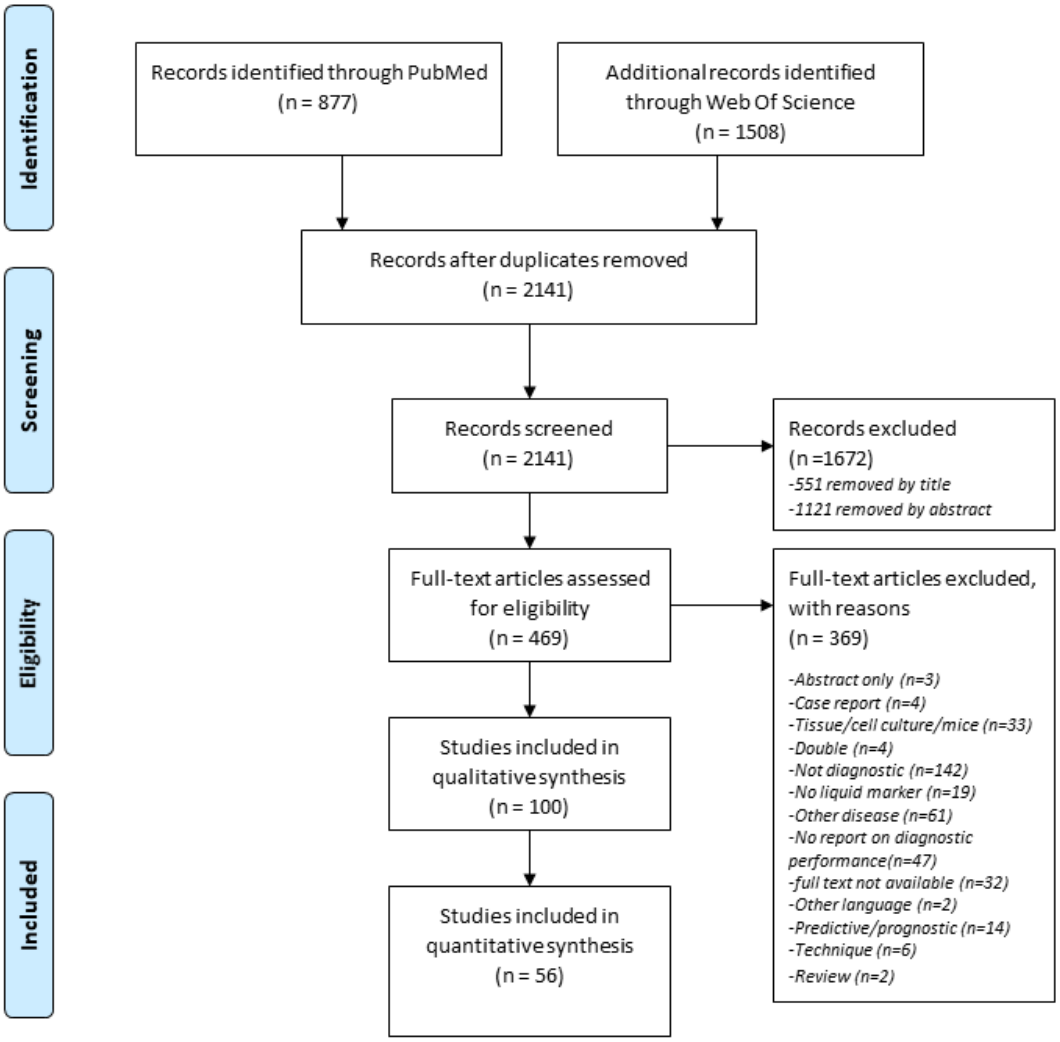
a. miRNA

Several miRNAs were studied individually by two research groups by the use of quantitative reverse transcriptase-polymerase chain reaction (RT-PCR) (7, 8). On their own, downregulation of miR-200c and upregulation of miR-21 resulted in the best AUC_{ROC} of 0.79.

b. DNA

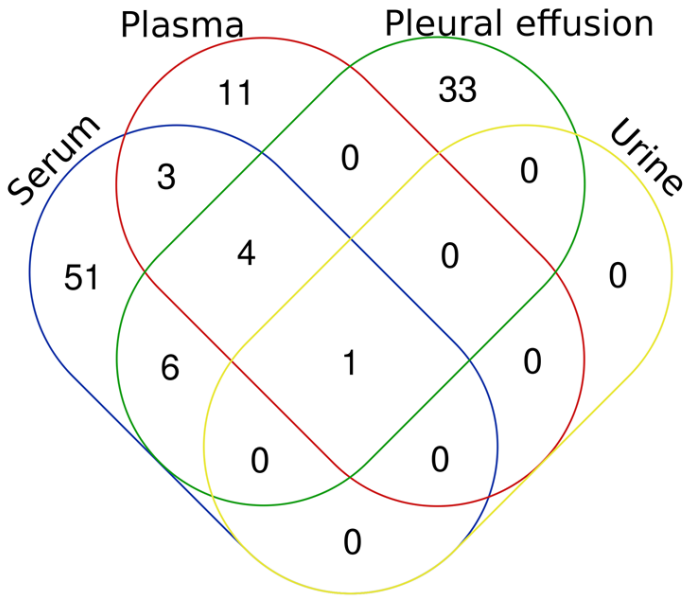
Two studies used RT-PCR to investigate the role of DNA as MPM biomarker in pleural effusions (9, 10). The methylation of RASSF1A, $p16^{INK4a}$ and RARβ were compared between MPM and LC patients, resulting in an differential diagnostic AUC_{ROC} of 0.752 at a methylation ratio of 0.05 in one of the three genes (9). When the DNA integrity index was studied in a validation cohort, an AUC_{ROC} of 0.82 was obtained discriminating MPM patients from those with benign diseases (10). A higher integrity index was observed in MPM patients and at a cut-off value of 1.02, 75% sensitivity and 90% specificity were reported.

Supplementary figure 1: *PRISMA flowchart of the study selection process.*



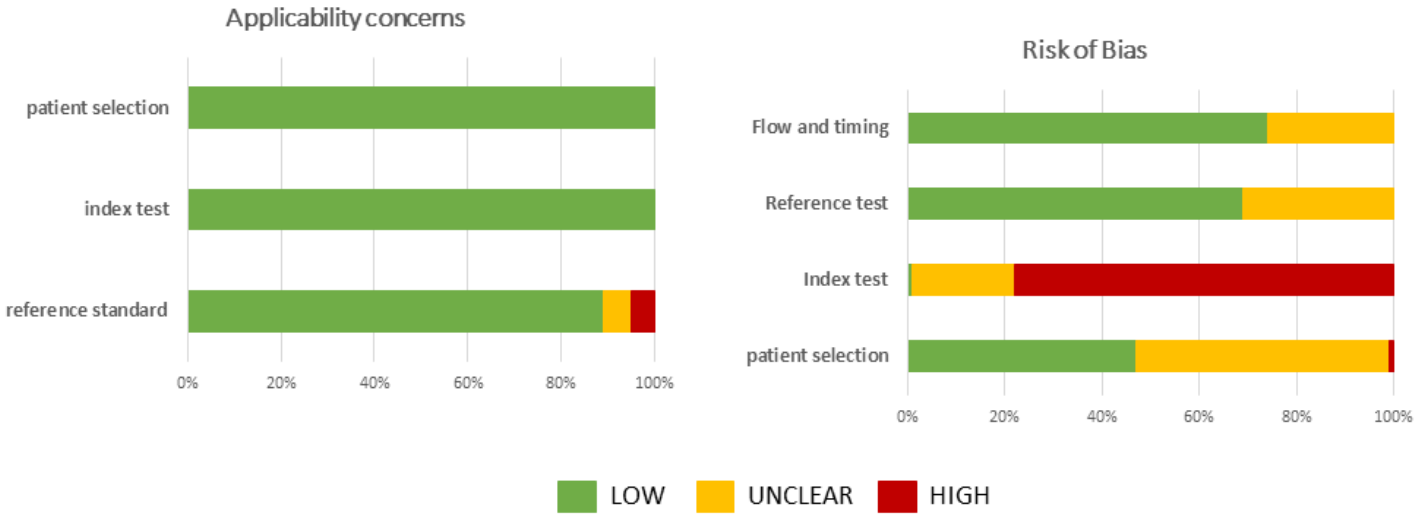
Supplementary figure 2: *Different biomarkers have been investigated in the different matrices. This venn-diagram shows the distribution of biomarkers per matrix. Serum is the most investigated matrix, followed by PE, plasma and urine. Mesothelin is the only marker that is investigated in all matrices.*

miR=micro RNA, MPF=megakaryocyte potentiating factor, SMRP=soluble mesothelin related peptides, OPN=osteopontin, CYFRA 21-1=cytokeratin-19 fragment, CEA=carcinoembryonic antigen, NSE=neuron specific enolase, HA=Hyaluronan, TSA= , CA15-3=cancer antigen 15-3

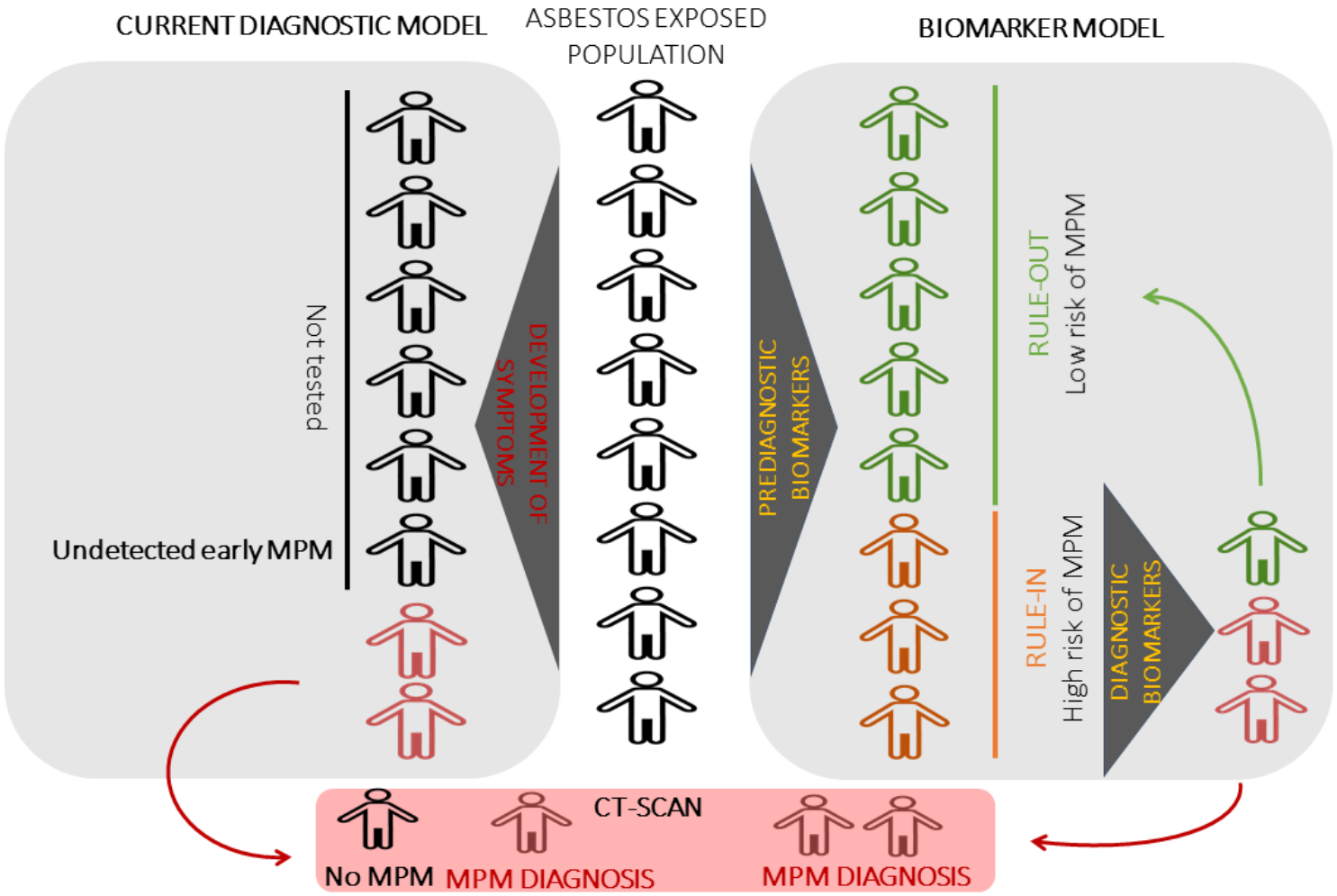


Serum|Plasma|Pleural effusion|Urine	*Mesothelin*
Serum|Plasma|Pleural effusion	*MPF, miR126, SMRP, OPN*
Serum|Plasma	*miR625-3p, Fibulin-3, Calretinin*
Serum|Pleural effusion	*CYFRA 21-1, CEA, NSE, HA, TSA, CA15-3*

Supplementary figure 3: *A summary of the quality assessment results using the QUADAS-2 tool.*



Supplementary figure 4: The role of (pre)diagnostic biomarkers in clinical practice. *The role of diagnostic biomarkers can be twofold. Firstly, they can be useful in a pre-diagnostic setting, where MPM can be detected earlier on, even before the onset of symptoms. Secondly, diagnostic biomarkers can reduce the amount of people unnecessarily exposed to radiation through CT-scans.*



AE=asbestos exposed, ARD=asbestos related disease, BPD= benign pleural disease, MPD= malignant pleural disease, MPM=malignant pleural mesothelioma

Supplementary table 1: *PECO strategy.*

POPULATION	*Patients with malignant pleural mesothelioma*
EXPOSURE	*Exposure to asbestos*
COMPARISON	*Healthy controls* *Patients with other malignant or benign asbestos related conditions* *Healthy asbestos exposed individuals*
OUTCOME	*Diagnostic accuracy of biomarkers in liquid biopsies for the discrimination between patients with MPM and controls (as described above)*

Supplementary table 2: *Overview of used search terms.*

PUBMED	*(((((((((mesothelioma[MeSH Terms]) OR pleural plaque OR asbestos[MeSH Terms]) OR asbestosis[MeSH Terms])) AND (((((((biomarkers[MeSH Terms]) OR chromosomes[MeSH Terms]) OR proteins[MeSH Terms]) OR genes[MeSH Terms]) OR RNA[MeSH Terms]) OR DNA[MeSH Terms]) OR peptides[MeSH Terms]) OR antibodies[MeSH Terms])) AND ((((((((((EBC) OR exhaled breath condensate) OR liquid biopsy[MeSH Terms]) OR blood[MeSH Terms]) OR serum[MeSH Terms]) OR plasma[MeSH Terms]) OR pleural effusion[MeSH Terms]) OR body fluid[MeSH Terms]) OR saliva[MeSH Terms]) OR urine[MeSH Terms])))))*
WEB OF SCIENCE	*TS=(mesothelioma OR asbestos OR asbestosis OR pleural plaque OR pleural plaques) AND TS=(biomarker OR biomarkers OR protein OR proteins OR RNA OR DNA OR peptide OR peptides OR gene OR genes OR chromosome OR chromosomes OR antibody OR antibodies) AND TS=(EBC OR exhaled breath condensate OR liquid biopsy OR liquid biopsies OR blood OR serum OR plasma OR pleural effusion OR pleural effusions OR urine OR saliva OR body fluid OR body fluids)*

Supplementary table 3: *An overview of studies reporting on MPM biomarkers in serum, plasma, pleural effusions and urine. When available the number of included epithelioid type MPM patients is reported between brackets.*

SERUM										
MARKER CATEGORY	**MARKER**	**STUDY DESIGN**	**METHOD**	**RESULTS**						**REFERENCE**
					AUC	**Cut-off**	**Sensitivity**	**Specificity**	**Expression in MPM**	
Protein	OPN	AE=69 HC=45 MPM=76	ELISA	MPM vs AE	0,888 (0,826-0,93)	48,3ng/ml	77,6%	85,5%	↑	(11)
	OPN	MPM=32 (32E) BARD=113 AE=94	ELISA	MPM vs BARD and AE	0,725	16,06ng/ml	62,5%	87,3%	↑	(12)
	OPN	MPM=24 PP=277 AE=123 HC=120	ELISA	MPM vs all other	0,86 (0,619-0,86)	17,273 nmol/L	75%	86%	↑	(13)
	Mesothelin				0,74 (0,773-0,951)	1,63 nmol/L	58%	83%	↑	
	Combination				-	OPN>17,273 or mesothelin >1,63	93%	73%	-	
	OPN	MPM=96 (73E) BARD=33 Metastase=43 AE=112	ELISA	MPM vs EA MPM vs BARD MPM vs metastase	0,724 (0,650-0,798) - -	-	-	-	↑ *(vs AE)*	(14)
	Mesothelin				0,866 (0,811-0,920) 0,834 (0,755-0,912) 0,719 (0,624-0,814)				↑	
	OPN	MPM=22 BPE=13 AE=20 Other cancer=25 HC=20	ELISA	MPM vs AE MPM vs BPE	0,502 (0,348-0,657) 0,502 (0,348-0,657)	-	-	-	~	(15)
	Mesothelin			MPM vs AE MPM vs BPE	0,934 (0,858-1,00) 0,798 (0,678-0,918)	-	-	-	↑	
	Fibulin-3			MPM vs AE MPM vs BPE	0,959 (0,905-1,000) 0,928 (0,868-0,989)	-	-	-	↑	
	HMGB1			*total and hyper acylated HMGB1* MPM vs AE MPM vs BPE MPM vs HC	0,830 (0,687-0,972) 1,000 (1,000-1,000)	15,75ng/ml 2,00ng/ml	100% 72,73%	5% 100%	↑	
					0,860 (0,736-0,984) 0,837 (0,709–0,966)	11,35ng/ml 9,70ng/ml	81,82% 77,27%	89,47% 89,47%		
					0,999 (0,994-1,000) 1,000(1,000-	-	-	-		

					1,000)					
				HMGB1 total and Fibulin-3 MPM vs BPE	0,987 (0.967–1.000)	-0,48	95,45%	92,11%	-	
				HMGB1 hyper acylated and Fibulin-3 MPM vs BPE	0.981 (0.956–1,000)	-0,22	90,91%	92,11%	-	
	HMGB1	MPM=15 AE=299 PP=81 Asbestosis=31 HC=71	ELISA	MPM vs HC MPM vs AE (<10y ago, n=170) MPM vs AE (>10y ago, n=129) MPM vs asbestosis	0,94 (0,89-0,99) 0,81 (0,73-0,90) 0,80 (0,72-0,89) 0,56 (0,39-0,73)	52,16ng/ml 52,29ng/ml 52,39ng/ml 51,64ng/ml	100% 100% 100% 100%	80,28% 57,65% 57,36% 29,03%	↑ *(excl. asbestosis)*	(16)
	MPF	MPM=66 (57E) AE=10 HC=10 ARD=21 BPE=30 MPE=20	ELISA	MPM vs HC and ARD	0,605 (0,499-0,711)	=specificity	39% 36% 34% 34%	85% 90% 95% 98%	↑ *(excl.LC, PE)*	(17)
	OPN				0,768 (0,679-0,857)		55% 53% 47% 22%	85% 90% 95% 98%	↑ *(excl.LC, PE)*	
	Mesothelin				0,905 (0,849-0,960)		81% 78% 73% 68%	85% 90% 95% 98%	↑	
	MPF,OPN, Mesothelin				0,920 (0,869-0,971)		-	-	-	
	Mesothelin, OPN				0,926 (0,878-0,974)				-	
	Mesothelin, MPF				0,916 (0,866-0,966)				-	
	OPN, MPF				0,745 (0,651-0,838)				-	
	Mesothelin	MPM=95 BPD=47 Other cancer=56	CLEIA ELISA	MPM vs other cancer MPM vs BAP MPM vs BPD MPM vs all other	0,68 (0,60-0,75) 0,78 (0,69-0,85) 0,73 (0,64-0,81) 0,71 (0,64-0,77)	1,5nmol/l	51,6% 51,6% 51,6% 51,6%	71,4% 85,0% 85,2% 77,7%	↑	(18)
	Midkine			MPM vs other cancer MPM vs BAP MPM vs BPD MPM vs all other	0,52 (0,44-0,60) 0,69 (0,60-0,78) 0,54 (0,44-0,63) 0,52 (0,45-0,59)	421pg/ml	61,1% 61,6% 61,6% 61,6%	41,1% 75,0% 48,1% 49,5%	~ (↑ *vs benign asbestos pleurisy)*	
	Combination			MPM vs other cancer MPM vs BAP MPM vs BPD MPM vs all other	0,59 (0,51-0,67) 0,83 (0,75-0,90) 0,65 (0,56-0,74) 0,73 (0,66-0,79)	-	88,4% 88,42% 100,0% 80;0%	32,7% 77,78% 0% 58,8%	-	
	Mesothelin	MPM=70 BARD=111 IPF=19 Sarcoidosis=19	ELISA	MPM vs BARD MPM vs benign disease	0,882 (0,826–0,925) 0,881 (0,832–0,920)	1,62nM	67%	95%	↑	(19)

		LC=20 BPE=7								
	N-ERC/mesothelin (MPF)	MPM=28 (21E) LC=11 Other cancer=2 BP=7 Undiagnosed=5	ELISA	MPM vs non MPM	0,74 0,77 (new ELISA)	8,14ng/ml 6,66ng/ml	64% 68%	72% 68%	↑	(20)
				Epithelioid vs non MPM	0,85 0,89 (new ELISA)	9,46ng/ml 9,16ng/ml	76% 81%	76% 80%		
	N-ERC/ Mesothelin (MPF)	MPM=91 (65E) Metastatic pleural disease=74 BARD=22 BPD=43	ELISA	MPM vs all other MPM vs metastatic MPM vs BARD MPM vs BPD	0,72 (0,65–0,77) 0,66 (0,58–0,73) 0,81 (0,72–0,88) 0,76 (0,67–0,83)	5.4ng/ml 5.4ng/ml 5.4ng/ml 5,6ng/ml	71,4% 71,4% 72,5% 67,4%	70,5% 60,8% 86,4% 83,7%	↑	(21)
	C-ERC/ Mesothelin			MPM vs all other MPM vs metastatic MPM vs BARD MPM vs BPD	0,69 (0,63–0,71) 0,64 (0,57–0,72) 0,76 (0,68–0,84) 0,74 (0,65–0,81)	162.9nM 164.7nM 180.9nM 167.5nM	67,0% 65,9% 59,3% 64,1%	66,2% 60,8% 86,4% 74,4%	↑	
	Hyaluronan			MPM vs all other MPM vs metastatic MPM vs BARD MPM vs BPD	0,58 (0,51–0,64) 0,55 (0,47–0,62) 0,70 (0,60–0,78) 0,57 (0,48–0,65)	37.9ng/ml 37.9ng/ml 35,0ng/ml 22,9ng/ml	74,7% 74,7% 74,7% 75,0%	44,6% 40,5% 63,6% 41,9%	*~ (↑ vs BAP)*	
	OPN			MPM vs all other MPM vs metastatic MPM vs BARD MPM vs BPD	0,56 (0,50–0,63) 0,62 (0,54–0,70) 0,56 (0,46–0,65) 0,52 (0,43–0,61)	139.1ng/ml 349.5ng/ml 216.1ng/ml 139.1ng/ml	31,9% 83,5% 46,2% 31,5%	79,9% 37,8% 77,3% 76,7%	~	
	Syndecan-1			MPM vs all other MPM vs metastatic MPM vs BARD MPM vs BPD	0,62 (0,55–0,68) 0,65 (0,57–0,72) 0,52 (0,42–0,62) 0,64 (0,55–0,72)	151.5ng/ml 151.5ng/ml 159.4ng/ml 151.5ng/ml	57,1% 57,1% 34,1% 57,6%	63,3% 67,6% 81,8% 65,1%	↓ (excl. BAP)	
	Mesothelin	LC=25 MPM=36 AE=26 ARD=48 PP=110 HC=109	ELISA	MPM vs AE	0,7359 (0,633-0,839)	1,2nM	61,11%	79,35%	↑	(22)
	Mesothelin	MPM=107 LC=215 BRD=130 HC=262	ELISA	MPM vs HC and BRD MPM vs LC	0,77 (0,71-0,83) 0,70 (0,63-0,77)	1nM 1nM	68,2% -	80,5% 68%	↑	(23)
	Mesothelin	MPM=100 (66E) Asbestosis=75 LC=139	ELISA	MPM vs ARD	0,72 (0,66-0,79)	2,3nM 1,35nM	33% 53%	95% 82,7%	↑	(24)
				MPM vs LC	0,719 (0,654-0,785)	2,5nM 1,25nM	30% 57%	95% 76,2%		
				MPM vs asbestosis	0,728 (0,655-0,802)	1,6nM 1,35nM	42% 53%	95% 88,0%		

	Mesothelin	HC=409 MPM=88 LC=174 Other cancer= Hypertension=100 Endometriosis=16	ELISA	MPM vs HC	0,874 (0,828–0,920)	1,9nmol/l 1,7 nmol/l **1,5 nmol/l** 1,3 nmol/l 1,0 nmol/l 0,9 nmol/l 0,8 nmol/l	43,2% 48,9% 56,8% 59,1% 68,2% 70,5% 75,0%	99,5% 99,3% 98,8% 97,1% 92,2% 87,5% 82,9%	↑	(25)
				MPM vs AE	0,806 (0,737–0,874)	1,9 nmol/l 1,7 nmol/l **1,5 nmol/l** 1,3 nmol/l 1,0 nmol/l 0,9 nmol/l 0,8 nmol/l	43,2% 48,9% 56,8% 59,1% 68,2% 70,5% 75,0%	100,0% 98,4% 95,1% 91,8% 77,0% 73,8% 68,9%		
				MPM vs all other	0,810 (0,755–0,865)	-	-	-		
	Mesothelin	MPM=52 (33E) Benign=43 Other cancer=25	ELISA	non positive cytology MPM(38) vs other	0,713 (0,603-0,822)	1,08nM	47,40%	82,50%	-	(26)
	Mesothelin	MPM=43 (26E) MPE=23 BPE=36	ELISA	MPM vs all other MPM vs BPE MPM vs MPE	0,705 (0,600-0,812) 0,729 (0,614-0,916) 0,739 (0,516-0,898)	1,08nM	46,5% 46,5% 46,5%	84,7% 91,7% 73,9%	↑	(27)
	Mesothelin	MPM=36 AE=177 BARD=101 HC=48	ELISA	MPM vs all other	0,753 50,65-0,84)	0,55 nmol/L	72%	72%	↑	(28)
	Mesothelin	MPM=139 Other cancer=30 HC=51	ELISA	MPM vs all other	0,89 (0,83–0,92)	-	-	-	↑	(29)
	Mesothelin	MPM=90 (58E) LC=170 AE=66	ELISA	MPM vs AE MPM vs LC MPM (stage I) vs AE MPM vs all other	0,810 (0,739-0,868) 0,820 (0,768-0,865) 0,741 (0,630-0,834) 0,891 (0,866-0,913)	1,9nM 1,1 nM 2,0nM 1,075nM	60% 78,9% 58% 81,1%	89,2% 76.4% 91% 87,1%	↑	(30)
	Mesothelin	MPM=60 BARD=23 Pleural metastases=35	ELISA	MPM vs BARD MPM vs metastases MPM vs all other	0,872 (0,798-0,946) 0,693 (0,576-0,810) 0,771 (0,683-0,858	0,93nM/l 1,85nM/l 1,1 nM/l	80% 58,3% 71,1%	82,6% 73,3% 69,8%	↑	(31)

	Mesothelin	MPM=48 AE=40 Lung inflammatory disease=92 Non-MPM PE=20 Benign, no PE=18 HC=28	ELISA	MPM vs other pleural disease MPM vs other lung tumours MPM vs AE	-	-	83%	100% 95.4% 82.5%	↑	(32)
	Mesothelin	MPM=27 (27E) AE=59	ELISA	MPM vs AE	-	1nM (homozygotes) 1,76nM(variant allele)	55,60% (both cut offs) 70,4% (1nM)	91,50% (both cut offs) 64,4% (1nM)	-	(33)
	Mesothelin	MPM=16 AE=293	ELISA	MPM vs AE	-	1,5nM	75,0%	96,2%	↑	(34)
	N-ERC/Mesothelin	MPM=39 PP=98 AE=83 LC=45 ARD=20 Other=8 HC=102	ELISA	MPM vs PP,AE, LC, ARD, other MPM vs PP, LC, BAP, other Epithelioid MPM vs PP, AE, ARD, LC, other Epithelioid MPM vs PP, BAP, LC, other	0,79 0,85 (new ELISA) 0,80 0,85 (new ELISA) 0,90 0,93 (new ELISA) 0,91 0,92 (new ELISA)	2,69ng/ml 5,60ng/ml 2,69ng/ml 5,60ng/ml 2,78ng/ml 5,60ng/ml 2,78ng/ml 5,60ng/ml	74,36% 71,79% 74,36% 71,79% 90,48% 90,48% 90,48% 90,48%	80,71% 93,31% 87,27% 92,12% 79,04% 90,44% 83,69% 87,98%	↑	(35)
	SMRP SNPs proximal MSLN promotor	HC=371 BRD=318 MPM=70	ELISA KASPar PCR SNP genotyping system	MPM vs all other	0,867 (0,841-0,890)	1,28nM 1nM	76,7% 87,7%	79,9% 64,1%	↑	(36)
				MPM vs all other (L)	0,922	1,11nM	83,6%	85,5%		
				MPM vs all other (H)	0,801	1,28nM	-	67%		
				MPM (L) vs all other (L)	0,914	1,11nM	79,5%	85,5%		
				MPM (H) vs all other (H)	0,829	1,28nM	-	67%		
	mesothelin	HC=101 AE=89 ARD=123 BRD=46 LC=63 MPM=85 (73E)	ELISA	MPM vs AE,ARD, LC MPM vs AE MPM vs ARD MPM vs LC	0,871 (0,825-0,916) 0,859 (0,811-0,908) 0,864 (0,805-0,924) 0,863 (0,806-0,921)	1,89nM 2,05nmol/l 2,03nmol/l 2,03nmol/L	66% 62% 64% 64%	94% 95% 95% 95%	↑	(37)
	MPF			MPM vs AE, ARD, LC MPM vs AE MPM vs ARD MPM vs LC	0,849 (0,792-0,906) 0,847 (0,789-0,905) 0,849 (0,786-0,913) 0,816 (0,747-0,886)	13,46ng/ml 12,66ng/ml 12,98ng/ml 13,46ng/ml	68% 68% 68% 68%	97% 95% 95% 95%	↑	

	Combination			MPM vs AE, ARD, LC	0,869 (0,820-0,919)	-	-	-	-	
	Mesothelin	MPM=41 (29E) AE=79	CLEIA	MPM vs AE	0,775 (0,682-0,868)	1,00µ/ml	74,7%	71,4%	↑	(38)
	sCD26				0,778 (0,690-0,865)	17,0µM/min	52,4%	82,3%	↓	
	DPPIV				0,738 (0,638-0,838)	-	-	-	↓	
	SMRP	MPM=45 (33E) AE=99 HC=44	ELISA Ref. gene: U6 small nuclear RNA, CelmiR-39 Method:TaqMan MicroRNA Assay	MPM vs AE and HC	0,818(0,723-O,914)	1nmol/l	60%	89%	↑	(39)
	miR126				0,710 (0,568-0,822)	$10x10^{-3}$	75%	54%	↓	
	Met-TM				0,750 (0,641-0,858)	1	60%	82%	↑	
					combination 0,857 (0,767-0,927)	-	-	-	↑	
	Mesothelin pOPN IL6 Vimentin	MPM=27 (27E) AE=43	ELISA MIA	MPM vs AE	0,962 (0,910–1,000)	0,13	100%	73%	-	(40)
	Calretinin	MPM=116 ARD=147	ELISA	MPM vs ARD	Nonparametric :0, 86 (0.82–0.91) bi-lognormal 0.90 (0.86–0.94)	1,07ng/ml 0,83ng/ml 0,68ng/ml 0,63ng/ml 0,58ng/ml	52% 60% 67% 68% 71%	99% 98% 97% 96% 95%	↑	(41)
	Mesothelin				Nonparametric 0,89 (0,85–0,93) bi-Weibull 0,91 (0,89–0,94)	2,32nmol/L 2,31ngmol/L 2,15nmol/L 2,02nmol/L 1,99nmol/L	61% 61% 66% 67% 69%	99% 98% 97% 96% 95%	↑	
	Mesothelin	MPM=117 (35E) ARD=53 BPE=30 AE=33	ELISA	MPM vs AE (with and without disease)	0,790 (0,729-0,851) 0,771 (0,710-0,832)	1,4nmol/L 1,7 nmol/L 2,2 nmol/L 2,5nmol/L	67% 60% 52% 49%	80% 90% 95% 98%	↑	(42)
	CA125				0,687(0,614-0,759) 0,610 (0,537-0,683)	6U/ml 8U/ml 10U/ml 25U/ml	52% 42% 27% 9%	80% 90% 95% 98%	↑	
	Combination				0,801 (0,741-0,861) 0,745 (0,682-0,807)	-	68% 60% 44% 42%	80% 90% 95% 98%	-	
	Mesothelin	MPM=93 (66E) LC=139 BARD=75	ELISA	MPM vs LC MPM vs BARD MPM vs LC and BARD	0,708 0,715 -	-	30%/55% 40%/51% 32%/55%	95% 95% 95%	↑	(43)
	CEA				0,011 0,16 -	-	-	-	↓	
	Combination				0,987	*Mesothelin/CEA*	93%/96%	95%/94%	-	

					0,887 -	*quotient* 0,9/0,83 (Youden optimization) 4,23/1,18 (Youden optimization) 2,37/1,18 (Youden optimization)	56%/89% 67%/89%	95%/72% 95%/88%		
	Mesothelin	MPM=31 (31E) AE=93 BRD=111	ELISA	MPM vs AE and BRD	0,762 (0,712-0,812)	0,675nM 0,605nM 0,575nM 0,555nM 1,01nM 1,14nM 1,32nM 1,56nM	80% 85% 90% 95% 61,3% 51,6% 51,6% 45,2%	50,5% 45,1% 40,7% 36,8% 80% 85% 90% 95%	↑	(44)
	Mesothelin + OPN (plasma)				0,873 (0,83-0,916)	0,165 0,153 0,044 0,039 0,106 0,117 0,156 0,251	80% 85% 90% 95% 83,9% 83,9% 80,6% 64,5%	91,2% 89,7% 45,1% 37,7% 80% 85% 90% 95%	-	
	Mesothelin	MPM=66 (59E) BARD=47 BPD=89 IPF=12 LC=21	ELISA	MPM vs benign	0,885 (0,831-0,939)	-	67%	95%	↑	(45)
	OPN				0,639 (0,545-0,733)		20%	95%	*~ (↑ vs benign, non astebstos related)*	
	Serum Mesothelin + plasma OPN				0,912 (0,866-0,959)		-	-	-	
	N-ERC + MPF	MPM=43 (28E) MPE=79 BPE=143	ELISA	MPM vs MPE, BPE	0,549 (0,465-0,630)	1,6ng/ml	95%	29%	~	(46)
	MPF				0,813 (0,741-0,872)	33,2ng/ml		52%	↑	
	Mesothelin				0,829 (0,759-0,886)	2,4nM		61%	↑	

	Mesothelin	MPM=106 HC=101 AE=215 ARD=78 LC=69 NRC=25	ELISA	MPM vs AE, ARD, LC, NRC	AGE 0.91 (0.88-0.95) 0.88 (0.84-0.92) 0.86 (0.81-0.91) 0.79 (0.73-0.85) GFR 0.80 (0.75-0.86) 0.87 (0.83-0.92) 0.88 (0.84-0.92) 0.92 (0.88-0.95) BMI 0.84 (0.78-0.89) 0.86 (0.81-0.91) 0.87 (0.83-0.92) 0.90 (0.86-0.94) STAGE 0.78 (0.69-0.86) 0.90 (0.86-0.94)	AGE (nmol/l) 1,71 1,77 2,02 2,99 GFR 3,34 2,00 1,78 1,71 BMI 2,51 2,81 1,88 1,67 STAGE 2,12 2,12	AGE 69% 67% 59% 39% GFR 36% 60% 67% 69% BMI 47% 41% 65% 69% STAGE 23% 69%	95%	↑	(47)
	MPF				AGE 0.87 (0.82-0.92) 0.85 (0.80-0.91) 0.84 (0.79-0.90) 0.78 (0.71-0.84) GFR 0.79 (0.73-0.85) 0.84 (0.79-0.90) 0.85 (0.80-0.91) 0.88 (0.83-0.93) BMI 0.82 (0.76-0.87) 0.84 (0.78-0.89) 0.84 (0.79-0.90) 0.86 (0.81-0.91) STAGE 0.76 (0.65-0.86) 0.89 (0.84-0.94)	AGE 11,92ng/ml 11,70 13,28 18,36 GFR 22,54 12,74 11,92 11,16 BMI 15,5 15,62 12,46 12,06 STAGE 13,42 13,42	AGE 67% 67% 66% 50% GRF 43% 66% 67% 68% BMI 60% 59% 66% 67% STAGE 46% 74%	95%	↑	

	Mesothelin	MPM=16 LC=22 Other cancer=8 BPE=23	ELISA RT qPCR	MPM vs benign	0,941 (0,869-1,00)	1,34nM	90%	75%	↑	(10)
	Mesothelin	MPM=24 Unspecified=158	ELISA	MPM vs other	0,61 (0,46–0,77)	1,5nM 0,975nM	59,3% 25,0%	64,7% 95%	↑	(48)
	Mesothelin	MPM=50 (32E) Asbestosis=16 NSCLC=77 HC=66	ELISA	MPM vs all other	0,85(0,79-0,91)	1,26nM	74%	86,1%	↑	(49)
	YKL-40				0,78(0,71-0,85)	156pg/ml	68%	74,3%	↑	
	Combination				0,86(0,80-0,91)	-	72%	83,8%	-	
	Mesothelin	MPM=42 HC=41 AE=48	ELISA	MPM vs AE, HC	0,86 (0,79-0,92)	0,62nmol/l	97,6%	68,9%	↑	(50)
	TRX				0,72(0,62-0,81)	156,67ng/ml	92,9%	77,6%	↑	
	EGFR				-	1,96ng/ml	90,5%	64,4%	↑	
	Fibulin-3				-	51,41ng/ml	88,1%	66,7%	↑	
	Syndecan-1				-	3,77ng/ml	90,0%	61,9%	↓	

	Mesothelin				-	2,04ng/ml	92,9%	61,1%	↑ *(~ vs AE)*	
	TRX	MPM=57 AE=15 BARD=19	ELISA	MPM vs ARD , AE	0,8178 (0,7482-0,8873)	60ng/ml	71,90%	85,00%	↑	(51)
	Mesothelin	MPM=39 LC=476 Other cancer=559 Benign=285 HC=147	ELISA	MPM vs HC MPM vs benign lung disease MPM vs LC MPM vs lung disease	0,902 0,771 0,751 0,755	-	95%/100%	71%/59% 45% 48% 48%	↑	(52)
	CYFRA21-1				0,928 0,817 0,610 0,639	-	95%/100%	75%/73% 46% 11% 15%	↑	
	NSE				0,869 0,733 0,615 0,635	-	95%/100%	66%/60% 32% 8% 8%	↑	
	CEA				0,440 0,274 0,119 0,145	-	95%/100%	8%/0% 0% 0% 0%	↓	
	SCCA				0,120 0,372 0,300 0,313	-	95%/100%	0%/0%/ 0% 0% 0%	↓	
	Mesothelin + CYFRA21-1				0,971 0,893 0,807 0,819	-	95%/100%	88%/88% 50% 46% 46%	-	
	VEGF	MPM=51 ARD=29 AE=13	ELISA	MPM vs ARD and AE	0,8623 (0,8001-0,9245)	460pg/ml	70,60%	88,10%	↑	(53)
	CA125	PP=277 AE=121 HC=118 MPM=24	ELISA	MPM vs AE, PP, HC	0,78	13,63nmol/L	83,3%	69,6%	↑	(54)
	CA15-3				0,68	18,43nmol/L	75%	48,4%	↑	
	Integrin-linked kinase	MPM=46 MPD=98 BPD=23	Western blotting	MPM vs MPD, BPD	0,891 (0,82-0,96)	2,48ng/ml	80%	95%	↑	(55)
	Integrin-linked kinase	MPM=101 AE=96	ELISA	MPM vs AE	0,69(0,63-0,76)	4,49ng/ml	61,4%	80,2%	↑	(56)
	CA 15-3	MPM=20 Benign=30 MPD=24	CLEIA ELISA colorimetric resorsinol method	MPM vs benign	-	20U/ml	80%	67%	↑	(57)
	CYFRA 21-1			MPM vs benign MPM vs bronchial cancer	-	4 ng/ml 10ng/ml	50% 79%	97% 65%	↑	
	CEA			MPM vs bronchial cancer	-	2ng/ml	83%	55%	~	
	NSE			MPM vs bronchial cancer	-	12µg/l	63%	70%	↑	

	TSA			MPM vs bronchial cancer	-	120mg/dl	75%	60%	↓	
	CEA	MPM=41 Other cancer=44	Immunoradiometricassay Competitive binding-protein radioimmunoassay enzyme immunoassay	MPM vs other cancer	0,807 (0,751-0,863)	-	-	-	↓	(58)
	CYFRA 21-1				0,594 (0,52-0,668)				↓	
	TPA				0,627 (0,562-0,692)				↓	
	13 protein marker algorithm (APOA1, C9, CCL23, CDK5/CDK5R1, CXCL13,F9,FCN2,FN1,ICAM2,KIT,MDK,SERPINA4, TNFRSF8)	MPM=38 AE=62	SOMA scan proteomic assay	MPM vs AE	0,95 +/-0,04	-	89,5% 93,2% (+mesothelin)	88,7% 90,8% (+mesothelin)	APOA1 ↓ C9 Up↑ CCL23 ↑ CDK5/CDK5 R1 ↑ CXCL13 ↑ F9 ↑ FCN2 ↑ FN1 ↓ ICAM2 ↑ KIT ↓ MDK ↑ SERPINA4 ↓ TNFRSF8 ↑	(59)
	Glycodelin	MPM=151 Pleurisy=23	ELISA	MPM vs benign	0,857 (0,819-0,896)	1,1ng/ml	81,5%	74,2%	↑	(60)
	PDIA6, MEG3, SDCCAG3, IGHG1, GHG3, NADH dehydrogenase 1, BACRP11-484D18, Clone CH507-528H12 on chromosome 21 andClone RP11-413M3 on chromosome 9	MPM=50 HC=30	phage display technology in combination with protein microarray	MPM vs HC *(validation)*	0,82	-	86,0%	86,0%	-	(61)
	17 antigens	MPM=35 AE=36	Luminex assay	MPM vs AE *(validation)*	Panel 1 0,51 (0,39-0,63) Panel 2 0,54 (0,42-0,66) Panel 3 0,50 (0,38-0,62) Panel 4 0,48 (0,36-0,630) Panel 5 0,48 (0,36-0,630) Panel 6 0,52 (0,40-0,64) Panel 70,54 (0,42-	ABL2 8157 RFU ALS2 5069 RFU BRWD1 5147 RFU IFIT3 1327 RFU KCNAB3 8939 RFU BC0104011 8899 RFU NME1 3720 RFU CCNE2 9878 RFU CDK2AP2 3849	-	-	-	(62)

					0,66) Panel 80,49 (0,37-0,60) Panel 9 0,46 (0,34-0,58) Panel 10 0,49 (0,37-0,60)	RFU KLC2 5282 RFU MBIP 9786 RFU RAB38 7062 RFU SPSB2 8638 RFU CSPP1 5392 RFU PP1R8 3112 RFU FUT10 1377 RFU ERRFI1 4066 RFU				
	Glycopeptides	UNK	electrospray ionization-liquid chromatography-mass spectrometry	MPM vs HC MPM vs NSCLC	*Training* 0,94 (0,87-0,99) 0,77 (0,62-0,91) *Validation* 0,94 (0,86-0,99) 0,56 (0,40-0,71) *+Mesothelin* 0,95 (0,83-1,0) 0,84 (0,66-0,97)	0,61 0,35	-	-	-	(63)
miRNA	miR-126	MPM=44 HC=50 AE=196	Ref. gene:- Method: TaqMan MicroRNA Assay	MPM vs AE	-	ΔCT -4,5	73%	74%	↓	(64)
	miR 197-3p	MPM=10 AE=10 HC=10	Ref. gene: miR-3665 Method: Human miRNA microarray G4470A, RT-qPCR	MPM vs HC MPM vs AE	0,718 (0,5398-0,8959) 0,763 (0,5981-0,9285)	-	-	-	↑	(65)
	miR-1281				0,759 (0,5885-0,9293) -				↑	
	miR-548-3p	MPM=60 AE=20 HC=20	Ref. gene: RNA RNU6 Method: miREasy RNA isolation kit, miScript SYBR Green PCR Kit	MPM vs AE and HC	0,922 (0,855-0,98)	1,69	91,7%	97,5%	↑	(66)
	miR-20a				0,98 (0,952-0,998)	1,2	96,7%	95%		
					0,96 (0,917-0,996) (combined)	1,7	100%	87,5%		
	miR-126	MPM=45 HC=56 NSCLC=20	Ref.gene: RNA U Method: Taq-Man miRNA reverse transcription kit	MPM vs HC MPM vs NSCLC	0,894 (0,821-0,968) 0,751 (0,6160,886)	-	-	-	↓	(67)
	miR 625-3p	MPM=30 Asbestosis=10	Ref. gene: mIR-16 Method: TaqMan MicroRNA Reverse Transcription Kit	MPM vs asbestosis	0,793 (0,657-0,930)	-	70%	90%	↑	(5)
	IncRNA-RP1	MPM=60 AE=20 HC=20	Ref. gene: Method: miRNeasy RNA isolation kit,RT-qPCR	MPM vs AE and HC	0,876 (0,802-0,950)	1,31	83,3%	95%	↑	(68)
	Hsa-1,3miR-2053				0,91 (0,848-0,971)	1,25	85%	97,5%	↑	
	DRAM1 mRNA				0,823 (0,735-0,911)	1,16	78,3%	87,5%	↓	

	ARSA mRNA				0,943(0,884-0,943)	1,52	90%	93,7%	↓	
	Combined				-	-	100%	85%	-	
	miR 34b/c methylation	MPM=48 ARD=21 HC=41	real time methylation specific PCR (MSP)assay	MPM vs ARD, HC	0,77	3mehtylated wells	67%	77%	↑	(69)
	miR 34b/c methylation	MPM=35 PP=29 HC=10	TaqMan-based ddPCR assay	MPM vs NM	0,85	fluorescence 7000/positive droplets 3	65,7%	94,9%	-	(70)
Lipids	31 lipids	MPM=40 AE=40 HC=40	LC high resolution MS	MPM vs AE, HC	L379.28 and L372.31 0,991 (0,976-1) L372.31 0,988 (0,967-1) L372.31 and L782.57 0,993 (0,981-1)	-	-	-	-	(2)
Cell (Blood)	CEC	MPM=79 BPD=30	CellTracks system Cell Track Analyzer II	MPM vs BPD	0,700 (0,595-0,806)	50 cells/4,0ml	49,4%	76,7%	↑	(4)
(blood)	CTC	MPM=104 NM=32	CellSearch system, Cell Track Analyzer II	MPM vs NM	0,623 (0,523-0,723)	1CTC	32,7%	90,6%	↑	(3)
PLASMA										
MARKER CATEGORY	**MARKER**	**STUDY DESIGN**	**METHOD**	**RESULTS**					**REFERENCE**	
					AUC	**Cut-off**	**Sensitivity**	**specificity**	**Expression in MPM**	
Protein	Mesothelin	MPM=75 AE=133 HC=107	ELISA	MPM vs HC (men/women)	0,90 (0,85-0,95) 0,92 (0,79-1,00)	-	-	-	↑	(71)
	Calretinin				0,88 (0,82-0,94) 0,77 (0,61-0,93)				↑	
	Trombomodulin				0,51 (0,41-0,61) 0,52 (0,32-0,72)				~	
	Mesothelin	MPM=166 (141E) Controls=378 (333=AE)	ELISA	MPM vs all other *-men* *-women*	0,914 (0,881–0,947) 0,913 (0,840–0,985)	1,457nmol/l 1,260nmol/l	70,3% 78,9%	95,3% 95,1%	↑	(72)
	Calretinin				0,923 (0,887–0,958) 0,893 (0,831–0,956)	0.330 ng/ml 0.671 ng/ml	81,1% 68,4%		↑	
	MPF				0,904 (0,868–0,940) 0,905 (0,839–0,970)	35.726 ng/ml 35,293 ng/ml	68,5% 63,2%		↑	
	Mesothelin + Calretinin + MPF				0,944 (0,916–0,973) 0,937 (0,876–	-	81,9% 86,8%		-	

					0,998)					
	Mesothelin + Calretinin				0,948 (0,921–0,974) 0,947 (0,897–0,997)	-	82,7% 86,8%		-	
	Mesothelin + MPF				0,920 (0,888–0,953) 0,928 (0,862–0,994)	-	71,7% 78,9%		-	
	Calretinin + MPF				0,938 (0,906–0,970) 0,919 (0,862–0,975)	-	84,3% 78,9%		-	
	Mesothelin	MPM=24 (15E) EA=136	ELISA	MPM vs AE	0,66 (0,55-0,78)	2,910 2,696 2,629 2,574 2,314 1,963	19% 19% 19% 19% 19% 29%	99% 98% 97% 96% 95% 90%	↑	(73)
	Calretinin				0,74 (0,63-0,85)	0,704 0,604 0,518 0,507 0,478 0,393	19% 26% 26% 26% 26% 42%	99% 98% 97% 96% 95% 90%	↑	
	Mesothelin and Calretinin				0,83 (0,65-0,83)	-	-	-	-	
	Mesothelin	MPM=82 (32E) BARD=49 NME=35 ME other cause=36	ELISA	MPM vs non MPM	0,816 (0,775-0,867)	2,5nM	56%	96%	↑	(74)
	Fibulin-3				0,671 (0,606-0,732)	52ng/ml(previously reported) 29ng/ml	22% 48%	95% 71%	↑	
	Combination				0,822 (0,762-0,872)	-	-	-	-	
	Mesothelin	MPM=66 (59E) BPD=89 BARD=47 IPF=12 LC=21	ELISA	MPM vs benign	0,864 (0,801-0,928)	-	76%	95%	↑	(45)
	OPN				0,763 (0,681-0,845)				↑	
	Serum mesothelin and plasma OPN				0,912 (0,866-0,959)				-	
	N-ERC mesothelin	MPM=28 (21E) LC=11 Other cancer=2 Benign pleuritis=7 Undiagnosed=5	ELISA	MPM vs non MPM *-normal ELISA* *-new ELISA* Epithelioid MPM vs non MPM *-normal ELISA* *-new ELISA*	0,74 0,82 0,86 0,91	6,14ng/ml 4,98ng/ml 7,62ng/ml 4,97ng/ml	71% 79% 81% 95%	72% 76% 76% 76%	↑ *(epithelioid)*	(20)

	Fibulin-3	MPM=92 AE=136 PE other cause=30 Benign PE=8 Malignant PE=22 No PE=259 *Validation:* MPM=48 AE=96	ELISA	MPM vs all other MPM vs AE MPM vs benign PE MPM vs malignant PE MPM vs PE other cause MPM vs no PE	0,99 (0,974-0,997) 0,99 (0,971-0,999) 0,95 (0,889-0,984) 0,94 (0,876-0,974) 0,94 (0,884-0,976) 0,99 (0,982-1,000)	52,8ng/ml 52,8 ng/ml 67,1 ng/ml 66,6 ng/ml 67,1 ng/ml 44,4 ng/ml	At 100%specificity 51,09% 71,32 82,61% 51,09% 51,09% 69,57%	At 100%sensitivity 74,48% 69,57% 25,00% 13,64% 16,67% 81,10%	↑	(75)
				Validation MPM vs AE	0,87	-	33%	100%		
	Fibulin-3	MPM=15 (15E) AE=21 Asbestosis=29 PP=74 HC=94	ELISA	MPM vs HC MPM vs AE MPM vs PP MPM vs Asbestosis	0,92 (0,81–1,00) 0,88 (0,75–1,00) 0,90 (0,78–1,00) 0,81 (0,67–0,96)	15.30 15,84 15,86 15,69	86,67% 86,67% 86,67% 86,67%	97,87% 91,89% 93,58% 75,86%	↑	(76)
	OPN	MPM=31 (31E) AE=93 BRD=48	ELISA	MPM vs AE and BRD	0,795 (0,745-0,845)	557,1ng/ml 428,2 ng/ml 336,9 ng/ml 334,5 ng/ml 782,1 ng/ml 902,7 ng/ml 1074,6 ng/ml 1423,9 ng/ml	80% 85% 90% 95% 71,0% 67,7% 58,1% 29,5%	68,1% 53,4% 31,9% 31,4% 80% 85% 90% 95%	↑	(44)
	OPN	MPM=32 (32E) BARD=113 AE=94	ELISA	MPM vs BARD and AE	0,78	878,65ng/ml	68,8%	84,5%	↑ *(epithelioid)*	(12)
	OPN	MPM=24 Chronic inflammatory disease=31 HC=37	ELISA	MPM vs HC MPM vs Chronic inflammatory disease Male MPM vs male HC	0,602 (0,47-0,724) 0,568(0,443-0,688) 0,719	60.8 ng/mL 12.2 ng/mL 60.8 ng/mL	40% 96% 58,8%	100% 25.8% 100%	~	(77)
	OPN	MPM=96 (73E) BARD=33 Metastases=43 AE=112	ELISA	MPM vs metastases MPM vs BARD	0,689 (0,579-0,798) 0,677 (0,562-0,792)	-	-	-	↓ *(vs metastases)*	(14)
	OPN	MPM=205 *(SMRP, n=139)*	ELISA	MPM vs all other	0,68 (0,61–0,74)	-	-	-	↑	(29)
	Mesothelin				0,89 (0,83–0,92)				↑	

		Other cancer=30 HC=51								
	Mesothelin	MPM=22 (11E) AE=94 HC=54	ELISA fluorometric OxyDNAassay kit *(8OHdG)*	MPM vs AE MPM vs HC	0,927 (0,905-0,949) 0,920 (0,89-0,95)	1,0 / 1,7 ng/ml 1,0 / 1,9 ng/ml	90% / 72% 90% / 68%	79% / 90% 78% / 90%	↑	(78)
	8OHdG				0,556 (0,446-0,666) 0,788 (0,698-0,878)	3,9 / 12,7 ng/ml 3,9 / 7,9 ng/ml	90% / 20% 90% / 60%	4% / 90% 37% / 90%	↑ *(~ vs AE)*	
	PDGFβ				0,765 (0,704-0,826) 0,783 (0,718-0,848)	4,3 /50,2 ng/ml 3,3 / 45,1 ng/ml	90% /45% 90% / 45%	38% / 90% 45% / 90%	↑	
	HGF				0,789 (0,726-0,852) 0,890 (0,841-0,939)	4,5 / 14,7 ng/ml 4,2 / 9,3 ng/ml	90% / 35% 90% / 75%	36% / 90% 48% / 90%	↑	
	bFGF				0,696 (0,623-0,769) 0,822 (0,761-0,883)	0,5 / 1,4 ng/ml 0,5 / 1,1 ng/ml	90% / 45% 90% / 50%	34% / 90% 52% / 90%	↑	
	VEGFβ				0,705 (0,619-0,791) 0,803 (0,729-0,877)	0,3 / 1,0 ng/ml 0,3 / 0,7 ng/ml	90% / 60% 90% / 70%	2% / 90% 17% / 90%	↑	
miRNA	miR 132-3p	Discovery: MPM=21 (14E) AE=21 Verification: MPM=22 (14E) EA=44	Ref gene: miR-146b-5p and U6 snRNA Method: TaqMan Low Density Array Human MicroRNA Card	MPM vs all other (Discovery/Verification) (sequential algorithm, and algorithm, or algorithm)	0,91 (0,80-1,00) 0,75 (0,63-0,88)	$2^{-(miR\ 132\text{-}3p\text{-}miR\ 146b)}$ 0,100 (YI max) 0,014 (FPR 0%) 0,047 (FPR 5%) 0,063 (FPR 10%)	86 5 23 36	61 100 95 90	↓	(79)
	miR 126 and miR 132-3p				0,88 (0,80-0,95) 0,84 (0,71-0,95) 0,77 (0,64-0,89)	$2^{-(miR\ 126\text{-}U6\text{-}snR))}$ *sequential algorithm* YI max 0,080/3,51 FPR 0% 0,020/0,10 FPR 5% 0,050/0,10 FPR 10% 0,060/0,62 *and algorithm* YI max 0,102/34,27 FPR 0% 0,132/0,62 FPR 5% 0,085/19,51	*sequential algorithm* 95 18 32 50 *and algorithm* 77 9 41 64	*sequential algorithm* 68 100 95 89 *and algorithm* 86 100 95 89	↓	

						FPR 10% 0,095/62,60				
	miR 17	MPM=32 Asbestosis=14 BRD=15	Ref. gene: miRNA-14 Method: TaqMan MicroRNA RT kit	MPM and ASB vs BRD *Sensibility specificity reported*	0,88 (0,78-0,98)	5.9	-	80.0–84.4	↓	(6)
	miR 126				0,95 (0,89-1,00)	5,4	-	80.0–97.8		
	miR486				0,88 (0,79-0,96)	9,2	-	80.0–89.1		
	miR16				0,89 (0,81-0,97)	77,5	-	86.7–82.2		
	miR-103	MPM=23 (12E) AE=17 HC=25	Ref gene: miR-125a Method: TaqMan miRNA Assay	MPM vs AE MPM vs HC	0,757 (0,586-0,929) 0.871 (0.766–0.977)	2-Ratio value of 0.621	83% 78%	71% 76%	↓	(80)
	miR 625-3p	MPM=15 HC (or with coronary disease)=14	Ref. gene: miR-16 Method: TaqMan MicroRNA Reverse Transcription Kit	MPM vs HC	0,824 (0,669-0,979)	-	73,33%	78,57%	↑	(5)
	Mesothelin	MPM=43 (28E) AE=52	ELISA	MPM vs AE *-All subtypes* *-Without sarcomatoid*	0,81 0,85	*1,70 nmol/l (max YI)* *2,01 nmol/l (FPR 4%)* *1,49 nmol/l (max YI)* *2,01 nmol/l (FPR 4%)*	*60* *49* *74* *55*	*92* *96* *85* *96*	↑	(81)
	miR-132-3p		Ref gene: miR-125a Method: TaqMan miRNA Assay		0,76 0,78	*749,61 (max YI)* *99,73 (FPR 4%)* *749,61 (max YI)* *99,73 (FPR 4%)*	*86* *7* *89* *8*	*63* *96* *63* *96*	↓	
	Combination		-		0,90 0,93	*-0,25≤-1,45-0,002*2-dCt+1,73*mesothelin (max YI)* *0,65≤-1,45-0,002*2-dCt+1,73*mesothelin (FPR 4%)* *-0,63≤-2,10-0,002*2-dCt+2,34*mesothelin (max YI)* *0,42≤-2,10-0,002*2-dCt+2,34*mesothelin (FPR 4%)*	*86* *65* *95* *74*	*85* *96* *81* *96*	-	
PLEURAL EFFUSION										
MARKER CATEGORY	MARKER	STUDY DESIGN	METHOD	RESULTS						REFERENCE
					AUC	Cut-off	Sensitivity	specificity	Expression in	

									MPM	
Protein	BDNF	MPM=86 BPE=18 Other neoplasia=85	ELISA	MPM vs other neoplasia MPM vs PE	0,6710 (0,5891-0,7529) 0,6972 (0,6223-0,7721)	20,16 20,16	49,51% 48,24%	86,05% 86,05%	↑	(82)
	Mesothelin	MPM=24 Other cancer=67 BPE=75	ELISA	MPM vs all other	0,878(0,807-0,948)	20nM	71%	90%	↑	(83)
	Mesothelin	*Discovery* MPM=46 Other cancer=49 Benign=95 *Validation* MPM=48 Other cancer=166 Benign=161	ELISA	MPM vs benign	0,885 (0,774-0,997)	12,91nM	90%	75%	↑	(10)
	Mesothelin	MPM=138 (59E) LC=182 Other cancer=253 Benign=712	ELISA	MPM vs all other	0,877 (0,843-0,91)	20nM	67%	95%	↑	(84)
	Mesothelin	MPM=24 Other=158	ELISA	MPM vs other	0,85 (0,77–0,94)	20nM 43,0nM	72,0% 50,0%	87,5% 95%	↑	(48)
	Mesothelin	MPM=52 (15E) Other cancer=56 Non-neoplastic=84	ELISA	MPM vs non-malignant MPM (excl. Sarcomatoid) vs non-malignant	0,898 0,964	20nM	67% 77%	98% 98%	↑	(85)
	Mesothelin	MPM=96 (73E) AE=112 BARD=33 Metastasis=43	ELISA	MPM vs BARD MPM vs Metastasis	0,829 (0,740-0,918) 0,758 (0,664-0,853)	-	-	-	↑	(14)
	Mesothelin	MPM=34 (25E) Metastasis=35 BPD=35	ELISA	MPM vs all other	0,767	19,6nM	58,8%	97,1%	↑	(86)
	Mesothelin	MPM=52 (35E) Other cancer=94 BPE=129	ELISA	MPM vs benign MPM vs other cancer MPM vs all other	0,840 (0,767-0,924) 0,800 (0,748-0,904) 0,820 (0,748-0,904)	*Max YI* 9,30nmol/l 9,30nmol/l 9,30nmol/l	75,0% 75,0% 75,0%	93,0% 80,9% 87,9%	↑	(87)
	Mesothelin	MPM=52 (33E) Metastasis=25 Benign=43	ELISA	MPM (not confirmed) vs all other	0,840 (0,749-0,930)	12,70nM	71,1%	87,7%	-	(26)
	Mesothelin	MPM=43 (26E) Metastases=23 Benign=36	ELISA	MPM vs all other MPM vs benign MPM vs metastases	0,816 (0,722-0,910) 0,815 (0,714-0,916) 0,817 (0,709-0,925)	12,70nM	69,8% 69,8% 69,8%	88.1% 91.7% 82.6%	↑	(27)
	Mesothelin	MPM=18	ELISA	MPM vs all other	0,7050 (0,5224–	6nmol/l	72%	46%	↑	(88)

		LC=10 Other cancer=20 ¨Pneumonia=4 TBC=4 Idiopatic=13			0,8875)	10nmol/l 30nmol/l 50nmol/l	61% 55% 44%	74% 92% 100%		
	Mesothelin	MPM=45 Benign=30 Other cancer=20	ELISA	MPM vs benign MPM vs all other	0,78 (0,664-0,863) 0,76 (0,662-0,842)	12,6nM 12,6nM	75,6% 75,6%	83,3% 82%	↑	(30)
	Mesothelin	MPM=43 Metastases=28 BARD=21	ELISA	MPM vs BARD MPM metastases MPM vs all other	0,831 (0,734-0,927) 0,793 (0,691-0,894) 0,809 (0,718-0,900)	10,4nM/l 11,4 - 36,8nM/l 11,4 - 20,8nM/l	76,7% 76% - 58,1% 76,7% - 65,1%	76,2% 64% - 93% 69,4% - 83,7%	↑	(31)
	Mesothelin	MPM=23 (15E) LC=38 BAP=26 TB=5 CHF=4	CLEIA	MPM vs all other MPM vs LC MPM vs BAP	0,75 (0,615-0,884) 0,724 (0,583-0,866) 0,74 (0,586-0,894)	8nmol/l	70% 69,6% 69,6%	68,4% 68,4% 69,2%	↑	(89)
	Fibulin-3	*Deteroit* MPM=78 AE=41 PE other cause=53 *New York* MPM=64 AE=95 *Validation* MPM=48 AE=96 PE other cause=40	ELISA	MPM vs all other *-Detroit* *-Ney York* MPM vs all PE MPM vs BPE MPM vs MPE	0,95/0,91 0,93 (0,880-0,966) 0,93(0,863-0,968) 0,94 (0,875-0,973)	378ng/ml/346 ng/ml 346,01ng/ml 378,33ng/ml 346,01ng/ml	- 83,8% 78,4% 83,8%	- 92,4% 94,9% 97,6%	↑	(75)
	Fibulin-3	MPM=82 (32E) BARD=49 BPE=35 MPE=36	ELISA	MPM vs all other	0,588 (0,505-0,668)	246ng/ml	59%	52%	~	(74)
	Mesothelin				0,815 (0,743-0,874)	20nM	58%	96%	↑	
	MPF	MPM=43 (28E) MPE=79 BPE=143	ELISA	MPM vs all other	0,945 (0,923-0,977)	600ng/ml	81%	95%	↑	(46)
	Mesothelin				0,928 (0,890-0,956)	20nM	70%	95%	↑	
	Hyaluronan	*Discovery* MPM=46 Other cancer=49 Benign=95 *Validation* MPM=48 Other cancer=166 Benign=161	High-performance LC	MPM vs all other	0,97 (0,94-1,00) 0,81 (0,75-0,88)	120µg UA/ml	100% 16,0%	37,9% 98,7%	↑	(90)
	N-ERC/mesothelin				0,81 (0,75-0,88) 0,81 (0,71-0,90)	1050ng/ml	100% 44,0%	87% 98,7%	↑	
	Combination				0,99 (0,97-1,00) 0,83 (0,74-0,91)	-	100% 52,0%	45,7% 98,7%	-	
	Fibulin-3	MPM=33 (22E) Metastases=23	ELISA	MPM vs benign MPM vs metastases	0,42 (0,28-0,55) 0,49 (0,33-0,65)	-	-	-	~	(91)

		Benign pleural lesions=64		MPM vs benign and metastases	0,44 (0,31-0,56)					
	Mesothelin				0,82 (0,72-0,92) 0,69 (0,54-0,83) 0,79 (0,68-0,89)	-	-	-	↑	
	CCL2	MPM=61 BPE=15 Metastases=25	ELISA	MPM vs metastases MPM vs BPE MPM vs metastasis and BPE	- - 0,7912 (0,7020–0,8803)	- - 1,61ng/ml	- - 72,50%	- - 80,33%	↑	(92)
	Galectine-3				- - 0,7584 (0,6652–0,8516)	- - 14,60ng/ml	- - 82,50%	- - 67,21%	↑ *(excl. Pulmonary ADCA)*	
	Mesothelin				0,902 (0,8385-0,9655) 0,912 (0,8433-0,9818) 0,9059 (0,8503–0,9616)	11,45nM 14,60nM 24,05nM	84,00% 93,33% 100,0%	81,97% 75,41% 63,93%	↑	
	SLPI				0,730 (0,6250-0,8365) 0,706 (0,5670-0,8450) 0,7219 (0,6207–0,8231)	168,20ng/ml 163,60ng/ml 168,20ng/ml	80,00% 80,00% 80,00%	70,49% 70,49% 70,49%	~	
	Combination				0,9680 (0,9120–0,9930)	-	-	-	-	
	CA153	MPM=18 ADCA=128 SCC=34 Other tumors=30 SCLC=17 Lymphoma/leukemia=13 Breast cancer=11	UniCel DxI 800 immunoassay system	MPM vs all other	0,635 (0,513-0,758)	-	-	-		(93)
	CA199				0,572 (0,437-0,707)	-	-	-		
	AFP				0,588 (0,451-0,724)	-	-	-		
	CEA				0,726 (0,593-0,858)	1,43ng/ml	83,7%	61,1%	↓ *(excl. Lymphoma/leukemia)*	
	CYFRA 21-1	MPM=33 LC=50 Other cancer=6 Malignant lymphoma=4 Benign disease=41	electrochemiluminescence immunoassay	MPM vs all other	-	69,7ng/ml 46,0ng/ml *(CEA<5ng/ml group)*	57,6% 62,5%	61,0% 88,0%	↑	(94)
	CYFRA 21-1/CEA				-	19,1	84,8%	80,2%	↑	
	CYFRA 21-1	MPM=32 Metastasis=40 Benign lung disease=34	ELISA	MPM vs benign	-	41,9ng/ml	87,5%	80%	↑	(95)
	CEA				-	5,0ng/ml	3,1%	91%	↓	
	CEA	MPM=41	IRMA	MPM vs other cancer	0,942 (0,910-	3ng/ml	100%	77%	↑	(58)

		Other cancer=44	ELISA		0,974)					
	HA				0,752 (0,687-0,817)	-	-	-	↑	
	CYFRA 21-1				0,654 (0,581-0,727)	-	-	-	↑	
	Mesothelin	MPM=57 (43E) Metastases=64 Benign=56	ELISA AXSYM CEA kit ECLEIA	MPM vs all other MPM vs metastases	0,84 (0,77-0,91) 0,81 (0,73-0,89)	12,0nmol/L 12,0nmol/l	74% 74%	86% 78%	↑	(96)
	CYFRA 21-1				0,76 (0,69-0,83) 0,76 (0,69-0,83)	71,5ng/l 100ng/l	91% 81%	60% 47%	↑	
	CEA				0,32 (0,24-0,40) 0,20 (0,12-0,28)	3,8ng/l 3,8ng/l	98% 98%	32% 56%	↑ (↓ *vs other malignancies)*	
	Mesothelin	MPM=32 (12E) PE other cause=208 (LC=44, other malignancy=17)	CLEIA	MPM vs PE other cause MPM vs other cancer	0,804 (0,690-0,918) 0,784 (0,679-0,889)	20nmol/l	56,3% 56,3%	86,5% 82,0%	↑	(97)
	CYFRA 21-1/CEA ratio				0,874 (0,780-0,948) 0,883 (0,805-0,961)	19,1	87,5% 87,5%	74,0% 82,0%	↑	
	Combination				-	-	93,9% 93,8%	64,9% 72,1%	-	
	CA 15-3	MPM=20 Other cancer=24 Benign disease=30	CLEIA ELISA colorimetric resorsinol method	MPM vs Benign MPM vs bronchial cancer	-	14U/ml 33U/ml	90% 63%	93% 70%	↑ *vs benign*	(57)
	CYFRA-21-1				-	8ng/ml -	90% -	90% -	↑ *vs benign*	
	CA 15-3 + CYFRA 21-1				-	-	100%	- 83%	-	
	CEA				-	- 3ng/ml	- 88%	- 90%	↑	
	CA 19-9				-	- 3U/ml	- 54%	- 70%	↑	
	NSE				-	- 3µg/l	- 63%	- 80%	~	
	TSA				-	- 80mg/dl	- 50%	- 80%	~	
	CEA+CA 15-3 + CA 19-9				-	-	- 92%	- 65%	-	
	CEA + CA 15-3 + TSA				-	-	- 96%	- 70%	-	
	CEA + CA 15-3				-	-	- 96%	- 70%	-	
	CEA + 15-3				-	-	- 96%	- 75%	-	

	CA 19-9, CA 15-3 And Kallikrein-12	MPM=30 TB=36 Lymphoma=12 ADCA=30 *Validation* MPM=20 TB=34 Lymphoma=18 ADCA=30	automated ultraviolet kinetic assay	MPM vs ADCA	0,94 (0,88-1)	>4 *CA19–9 13.3 U/mL (3 points), kallikrein-12 <55 ng/mL (2 points), and CA15-3 >110 U/mL (1 point)*	65%	100%	-	(98)
	HE4	MPM=41 (21E) PE other cause=48	ELISA SELDI-TOF MS	MPM vs PE other cause	0,440		-	-	~	(99)
	OPN				0,592		-	-	~	
	CYFRA 21-1				0,736 (0,630-0,841)	-	-	-	↑	
	Mesothelin				0,860 (0,782-0,938)	7,05nM/l 14nM/l	83% 73%	74% 80%	↑	
	Protein peak 6614 Da				0,741 (0,636-0,845)	m/z 132	76%	69%	↓	
	Protein peak 6626Da				0,755	-	-	-	↓	
	Protein peak 8799D				0,688	-	-	-	↓	
	Protein peak 6656Da				0,728	-	-	-	↓	
	Protein peak 6821Da				0,742	-	-	-	↓	
	Aldo-keto reductase 1B10	*Screening* MPM=6 LC=6 Benign=7 *Validation* MPM=37 (30E) Metastases=25 Benign=16	Liquid Chromatography Electrospray Ionization LTQ-Orbitrap Analysis ELISA	MPM vs non-MPM MPM vs metastases	0,71 (0,59–0,83) 0,66 (0,52–0,81)	-	-	-	↑	(100)
	Apolipoprotein C-I				0,60 (0,46–0,73) 0,57 (0,41–0,74)				↑	
	Galectin 1				0,87 (0,79–0,95) 0,93 (0,86–0,99)				↑	
	Myosin-VIIb				0,60 (0,47–0,74) 0,59 (0,43–0,75)				↑	
	Superoxide dismutase 2				0,64 (0,51–0,78) 0.71 (0,55–0,86)				↑	
	Tenascin C				0,54 (0,40–0,68) 0,56 (0,40–0,72)				↑	
	Thrombospondin 1				0,54 (0,40–0,68) 0,54 (0,37–0,70)				↑	
	Hyaluronan				0,71 (0,59–0,83) 0.82 (0,70–0,94)				↑	
	N-ERC /mesothelin				0,89 (0,80–0,97) 0,87 (0,77–0,96)				↑	
	sCD44 std	MPM=33 Metastasis=65	ELISA	MPM vs Other cancer MPM vs ADCA	-	-	-	-	~	(101)
	sCD44 v6				-	100ng/ml	52%	87%	↓	

		Lymphoma=39 BPE=24		MPM vs Other cancer and ADCA	- 0,80 (0,71-0,91)		60% -	87% -		
	sCD44 v6/std ratio				- - 0,87 (0,80-0,94)	0,34	50% 60% -	93% 93% -	↓	
DNA	RASSF1A,p16^{INK4a} and RARβ	MPM=39 LC=46 BAP=25 Other pulmonary diseases=30	Ref. gene: β-actin Method: Taqman rtPCR	MPM vs LC	0,752 (0,661-0,843)	0,05 methylation ratio *(in one of three genes)*	67,4%	79,5%	↓ *(vs LC)*	(9)
	DNA integrity index	*Discovery* MPM=46 Other cancer=49 Benign=95 *Validation* MPM=48 Other cancer=166 Benign=161	RT qPCR	MPM vs benign	0,823 (0.676-0.970)	1,02	75%	90%	↑ *(vs benign)*	(10)
miRNA	miR 210	MPM=26 ADCA=11 Benign=10	Ref. gene: mesothelin and fibulin-3 mRNA Method: miRNeasy Kit, Taq-Man Reverse Transcription Kit	MPM vs all other MPM vs ADCA	0,72 (0,58-0,8) -	-	-	-	↑	(7)
	miR 143				0,66 (0,50-0,82) -	-	-	-	↓	
	miR 200c				0,79 (0,66-0,92) -	-	-	-	↓	
	miR 139-5p				0,65 (0,50-0,8) -	-	-	-	↑	
	miR 200c, miR 210 and miR 143				0,92 (0,84-0,99) 0,99(0,97-1,00)	-	- 92.31%	- 96.08%	-	
	miR-19a	MPM=29 (29E) RMC=24	Ref. gene: RNA U6B Method: qRT-PCR	MPM vs RMC	0,69 (0,54-0,83)	2,64	88%	52%	↑	(8)
	miR-19b				0,68 (0,54-0,83)	0,03	90%	90%	↑	
	miR-21				0,79 (0,66-0,91)	20,87	45%	45%	↑	
	miR-25				0,59 (0,43-0,75)	0,16	72%	72%	~	
	miR-126				0,71 (0,57-0,85)	0,12	52%	52%	↓	
URINE										
MARKER CATEGORY	MARKER	STUDY DESIGN	METHOD	RESULTS						REFERENCE
					AUC	Cut-off	Sensitivity	specificity	Expression in MPM	
Protein	Mesothelin	MPM=70 ARD=111 IPF=19 LC=20 Sarcoidosis=19 BPE=7	ELISA	MPM vs BPD	0,870 (0,764–0,940)	-	60%	95%	↑	(19)

ADCA= adenocarcinoma, AE=asbestos exposed, AFP=alpha-fetoprotein, ARD=asbestos related disease, BAP=benign asbestos pleurisy, BARD=benign asbestos related disease, BDNF=brain-derived neurotrophic factor, bFGF=basic fibroblast growth factor, BMI=body mass index, BP=benign pleurisy, BPD=benign pleural disease, BPE=benign pleural effusion, BRD=benign respiratory disease, CCL2=C-C motif chemokine ligand 2, CEA=carcinoembryonic antigen,, CEC=circulating endothelial cells,

CHF=chronic heart failure, CLEIA= chemiluminescent enzyme immunoassay, CTC=circulating tumor cells, CYFRA=cytokeratin fragment, DPPIV=dipeptidyl peptidase-4, ECLEIA=electro-chemiluminescence immunoassay, EGFR=epidermal growth factor receptor, ELISA=enzyme-linked immunosorbent assay, GFR=glomerular filtration rate, HC=healthy control, HA=hyaluronan, HC=healthy control, HE4=human epididymis protein, HGF=hepatocyte growth factor, HMGB=high mobility group B, IL=interleukine, IPF=idiopathic pulmonary fibrosis, IRMA=immunoradiometric assay, KASPar PCR SNP=kompetitive allele specific polymerase chain reaction single nucleotide polymorphism, LC=lung cancer, (high performance) LC=liquid chromatography, ME=malignant effusion, Met-TM=methylated thrombomodulin promotor, MIA=magnetic immunoassay, miR=micro RNA, MPD=malignant pleural disease, MPE=malignant pleural effusion, MPF=megakaryocyte potentiating factor, MPM=malignant pleural mesothelioma, MSLN=mesothelin, NM=non-malignant, NME=non-mesothelioma effusion, NRC=non respiratory cancer, NSE=neuron-specific enolase, NSCLC=non-small cell lung cancer, OHdG=hydroxyguanosine, OPN=osteopontin, PBD=benign pleural disease, PDGF=platelet-derived growth factor, PE=pleural effusion, PP=pleural plaque, RMC= reactive mesothelial cells, RT qPCR=real time quantitative polymerase chain reaction, SCC=squamous cell carcinoma, SCCA=squamous cell cancer antigen, SCLC= small cell lung cancer, SELDI-TOF MS=surface-enhanced laser desorption/ionization time-of-flight mass spectrometry, SLPI= secretory leukocyte protease inhibitor, SMRP=soluble mesothelin related peptide, TBC=tuberculosis, TPA=tissue plasminogen activator, TRX=thioredoxin-1, TSA=total sialic acid, VEGF=vascular endothelial growth factor, YKL=chitinase-3 like protein 1.

REFERENCES

1. Yan F, Zhao H, Zeng Y. Lipidomics: a promising cancer biomarker. *Clin Transl Med* 2018; 7: 21-21.
2. Xu RY, Mesaros C, Weng LW, Snyder NW, Vachani A, Blair IA, Hwang WT. Comparison of statistical methods for detection of serum lipid biomarkers for mesothelioma and asbestos exposure. *Biomark Med* 2017; 11: 547-556.
3. Yoneda K, Tanaka F, Kondo N, Hashimoto M, Takuwa T, Matsumoto S, Okumura Y, Tsubota N, Sato A, Tsujimura T, Kuribayashi K, Fukuoka K, Tabata C, Nakano T, Hasegawa S. Circulating Tumor Cells (CTCs) in Malignant Pleural Mesothelioma (MPM). *Ann Surg Oncol* 2014; 21: S472-S480.
4. Yoneda K, Tanaka F, Kondo N, Orui H, Hashimoto M, Takuwa T, Matsumoto S, Okumura Y, Tsubota N, Sato A, Tsujimura T, Kuribayashi K, Fukuoka K, Nakano T, Hasegawa S. Circulating Endothelial Cell (CEC) as a Diagnostic and Prognostic Marker in Malignant Pleural Mesothelioma (MPM). *Ann Surg Oncol* 2012; 19: 4229-4237.
5. Kirschner MB, Cheng YY, Badrian B, Kao SC, Creaney J, Edelman JJB, Armstrong NJ, Vallely MP, Musk AW, Robinson BWS, McCaughan BC, Klebe S, Mutsaers SE, van Zandwijk N, Reid G. Increased Circulating miR-625-3p A Potential Biomarker for Patients With Malignant Pleural Mesothelioma. *J Thorac Oncol* 2012; 7: 1184-1191.
6. Mozzoni P, Ampollini L, Goldoni M, Alinovi R, Tiseo M, Gnetti L, Carbognani P, Rusca M, Mutti A, Percesepe A, Corradi M. MicroRNA Expression in Malignant Pleural Mesothelioma and Asbestosis: A Pilot Study. *Dis Markers* 2017: 10.
7. Birnie KA, Prele CM, Musk AW, de Klerk N, Lee YCG, Fitzgerald D, Allcock RJN, Thompson PJ, Creaney J, Badrian B, Mutsaers SE. MicroRNA Signatures in Malignant Pleural Mesothelioma Effusions. *Dis Markers* 2019; 2019: 9.
8. Cappellesso R, Nicole L, Caroccia B, Guzzardo V, Ventura L, Fassan M, Fassina A. Young investigator challenge: MicroRNA-21/MicroRNA-126 profiling as a novel tool for the diagnosis of malignant mesothelioma in pleural effusion cytology. *Cancer Cytopathol* 2016; 124: 28-37.
9. Fujii M, Fujimoto N, Hiraki A, Gemba K, Aoe K, Umemura S, Katayama H, Takigawa N, Kiura K, Tanimoto M, Kishimoto T. Aberrant DNA methylation profile in pleural fluid for differential diagnosis of malignant pleural mesothelioma. *Cancer Sci* 2012; 103: 510-514.
10. Sriram KB, Relan V, Clarke BE, Duhig EE, Windsor MN, Matar KS, Naidoo R, Passmore L, McCaul E, Courtney D, Yang IA, Bowman RV, Fong KM. Pleural fluid cell-free DNA integrity index to identify cytologically negative malignant pleural effusions including mesotheliomas. *BMC Cancer* 2012; 12: 428.
11. Pass HI, Lott D, Lonardo F, Harbut M, Liu ZD, Tang NM, Carbone M, Webb C, Wali A. Asbestos exposure, pleural mesothelioma, and serum osteopontin levels. *New England Journal of Medicine* 2005; 353: 1564-1573.
12. Cristaudo A, Foddis R, Bonotti A, Simonini S, Vivaldi A, Guglielmi G, Ambrosino N, Canessa PA, Chella A, Lucchi M, Mussi A, Mutti L. Comparison between plasma and serum osteopontin levels: usefulness in diagnosis of epithelial malignant pleural mesothelioma. *Int J Biol Markers* 2010; 25: 164-170.
13. Bayram M, Dongel I, Akbas A, Benli I, Akkoyunlu ME, Bakan ND. Serum Biomarkers in Patients with Mesothelioma and Pleural Plaques and Healthy Subjects Exposed to Naturally Occurring Asbestos. *Lung* 2014; 192: 197-203.
14. Grigoriu BD, Scherpereel A, Devos P, Chahine B, Letourneux M, Lebailly P, Gregoire M, Porte H, Copin MC, Lassalle P. Utility of osteopontin and serum mesothelin in malignant pleural mesothelioma diagnosis and prognosis assessment. *Clin Cancer Res* 2007; 13: 2928-2935.
15. Napolitano A, Antoine DJ, Pellegrini L, Baumann F, Pagano I, Pastorino S, Goparaju CM, Prokrym K, Canino C, Pass HI, Carbone M, Yang HN. HMGB1 and Its Hyperacetylated Isoform are Sensitive and Specific Serum Biomarkers to Detect Asbestos Exposure and to Identify Mesothelioma Patients. *Clin Cancer Res* 2016; 22: 3087-3096.

16. Ying SB, Jiang ZQ, He XL, Yu M, Chen RP, Chen JQ, Ru GQ, Chen Y, Chen WY, Zhu LJ, Li T, Zhang YX, Guo XN, Yin XH, Zhang X, Lou JL. Serum HMGB1 as a Potential Biomarker for Patients with Asbestos-Related Diseases. *Dis Markers* 2017: 9.
17. Creaney J, Yeoman D, Demelker Y, Segal A, Musk AW, Skates SJ, Robinson BWS. Comparison of osteopontin, megakaryocyte potentiating factor, and mesothelin proteins as markers in the serum of patients with malignant mesothelioma. *J Thorac Oncol* 2008; 3: 851-857.
18. Ak G, Tada Y, Shimada H, Metintas S, Ito M, Hiroshima K, Tagawa M, Metintas M. Midkine is a potential novel marker for malignant mesothelioma with different prognostic and diagnostic values from mesothelin. *Bmc Cancer* 2017; 17: 10.
19. Creaney J, Musk AW, Robinson BWS. Sensitivity of Urinary Mesothelin in Patients with Malignant Mesothelioma. *J Thorac Oncol* 2010; 5: 1461-1466.
20. Sato T, Suzuki Y, Mori T, Maeda M, Abe M, Hino O, Takahashi K. Newly established ELISA for N-ERC/mesothelin improves diagnostic accuracy in patients with suspected pleural mesothelioma. *Cancer Med* 2014; 3: 1377-1384.
21. Arslan S, Mundt F, Metintas S, Ak G, Dobra K, Hjerpe A, Metintas M. Predicting Malignant Mesothelioma by Analyzing Serum N-ERC/Mesothelin, C-ERC/Mesothelin, Hyaluronan, Osteopontin, and Syndecan-1 Levels. *Eurasian J Pulmonol* 2017; 19: 130-138.
22. Dipalma N, Luisi V, Di Serio F, Fontana A, Maggiolini P, Licchelli B, Mera E, Bisceglia L, Galise I, Loizzi M, Pizzigallo MA, Molinini R, Vimercati L. Biomarkers in malignant mesothelioma: diagnostic and prognostic role of soluble mesothelin-related peptide. *Int J Biol Markers* 2011; 26: 160-165.
23. Cristaudo A, Foddis R, Vivaldi A, Guglielmi G, Dipalma N, Filiberti R, Neri M, Ceppi M, Uzzi MP, Ivaldi GP, Mencoboni M, Canessa P, Ambrosino N, Chella A, Mutti L, Puntoni R. Clinical significance of serum mesothelin in patients with mesothelioma and lung cancer. *Clin Cancer Res* 2007; 13: 5076-5081.
24. Schneider J, Hoffmann H, Dienemann H, Herth FJF, Meister M, Muley T. Diagnostic and Prognostic Value of Soluble Mesothelin-Related Proteins in Patients with Malignant Pleural Mesothelioma in Comparison with Benign Asbestosis and Lung Cancer. *J Thorac Oncol* 2008; 3: 1317-1324.
25. Beyer HL, Geschwindt RD, Glover CL, Tran L, Hellstrom I, Hellstrom KE, Miller MC, Verch T, Allard WJ, Pass HI, Sardesai NY. MESOMARK (TM): A potential test for malignant pleural mesothelioma. *Clin Chem* 2007; 53: 666-672.
26. Franceschini MC, Ferro P, Canessa PA, Battolla E, Dessanti P, Valentino A, Casolari L, Fontana V, Pezzi R, Fedeli F, Pistillo MP, Roncella S. Mesothelin in serum and pleural effusion in the diagnosis of malignant pleural mesothelioma with non-positive cytology. *Anticancer Res* 2014; 34: 7425-7429.
27. Ferro P, Canessa PA, Battolla E, Dessanti P, Franceschini MC, Chiaffi L, Morabito A, Fontana V, Pezzi R, Fedeli F, Pistillo MP, Roncella S. Mesothelin is more useful in pleural effusion than in serum in the diagnosis of pleural mesothelioma. *Anticancer Res* 2013; 33: 2707-2713.
28. Portal JAR, Becerra ER, Rodriguez DR, Michavila IA, Martinez AQ, Roza CD, Jimenez AL, Montes, II, Rivas PC. Serum Levels of Soluble Mesothelin-Related Peptides in Malignant and Nonmalignant Asbestos-Related Pleural Disease: Relation with Past Asbestos Exposure. *Cancer Epidemiol Biomarkers Prev* 2009; 18: 646-650.
29. Rai AJ, Flores RM, Mathew A, Gonzalez-Espinoza R, Bott M, Ladanyi M, Rusch V, Fleisher M. Soluble mesothelin related peptides (SMRP) and osteopontin as protein biomarkers for malignant mesothelioma: analytical validation of ELISA based assays and characterization at mRNA and protein levels. *Clin Chem Lab Med* 2010; 48: 271-278.
30. Pass HI, Wali A, Tang N, Ivanova A, Ivanov S, Harbut M, Carbone M, Allard J. Soluble mesothelin-related peptide level elevation in mesothelioma serum and pleural effusions. *Ann Thorac Surg* 2008; 85: 265-272; discussion 272.
31. Scherpereel A, Grigoriu B, Conti M, Gey T, Gregoire M, Copin MC, Devos P, Chahine B, Porte H, Lassalle P. Soluble mesothelin-related peptides in the diagnosis of malignant pleural mesothelioma. *Am J Respir Crit Care Med* 2006; 173: 1155-1160.

32. Robinson BWS, Creaney J, Lake R, Nowak A, Musk AW, de Klerk N, Winzell P, Hellstrom KE, Hellstrom I. Soluble mesothelin-related protein - A blood test for mesothelioma. *Lung Cancer* 2005; 49: S109-S111.
33. Cristaudo A, Foddis R, Bonotti A, Simonini S, Vivaldi A, Guglielmi G, Bruno R, Gemignani F, Landi S. Two novel polymorphisms in 5 ' flanking region of the mesothelin gene are associated with soluble mesothelin-related peptide (SMRP) levels. *Int J Biol Markers* 2011; 26: 117-123.
34. Smolkova P, Nakladalova M, Zapletalova J, Jakubec P, Vildova H, Kolek V, Petrek M, Nakladal Z. VALIDITY OF MESOTHELIN IN OCCUPATIONAL MEDICINE PRACTICE. *Int J Occup Med Environ Health* 2016; 29: 395-404.
35. Shiomi K, Hagiwara Y, Sonoue K, Segawa T, Miyashita K, Maeda M, Izumi H, Masuda K, Hirabayashi M, Moroboshi T, Yoshiyama T, Ishida A, Natori Y, Inoue A, Kobayashi M, Sakao Y, Miyamoto H, Takahashi K, Hino O. Sensitive and specific new enzyme-linked immunosorbent assay for N-ERC/Mesothelin increases its potential as a useful serum tumor marker for mesothelioma. *Clin Cancer Res* 2008; 14: 1431-1437.
36. De Santi C, Pucci P, Bonotti A, Melaiu O, Cipollini M, Silvestri R, Vymetalkova V, Barone E, Paolicchi E, Corrado A, Lepori I, Dell'Anno I, Pelle L, Vodicka P, Mutti L, Foddis R, Cristaudo A, Gemignani F, Landi S. Mesothelin promoter variants are associated with increased soluble mesothelin-related peptide levels in asbestos-exposed individuals. *Occup Environ Med* 2017; 74: 457-464.
37. Hollevoet K, Nackaerts K, Thimpont J, Germonpre P, Bosquee L, De Vuyst P, Legrand C, Kellen E, Kishi Y, Delanghe JR, van Meerbeeck JP. Diagnostic Performance of Soluble Mesothelin and Megakaryocyte Potentiating Factor in Mesothelioma. *Am J Respir Crit Care Med* 2010; 181: 620-625.
38. Fujimoto N, Ohnuma K, Aoe K, Hosono O, Yamada T, Kishimoto T, Morimoto C. Clinical Significance of Soluble CD26 in Malignant Pleural Mesothelioma. *Plos One* 2014; 9: 14.
39. Santarelli L, Staffolani S, Strafella E, Nocchi L, Manzella N, Grossi P, Bracci M, Pignotti E, Alleva R, Borghi B, Pompili C, Sabbatini A, Rubini C, Zuccatosta L, Bichisecchi E, Valentino M, Horwood K, Comar M, Bovenzi M, Dong LF, Neuzil J, Amati M, Tomasetti M. Combined circulating epigenetic markers to improve mesothelin performance in the diagnosis of malignant mesothelioma. *Lung Cancer* 2015; 90: 457-464.
40. Bonotti A, Foddis R, Landi S, Melaiu O, De Santi C, Giusti L, Donadio E, Ciregia F, Mazzoni MR, Lucacchini A, Bovenzi M, Comar M, Pantani E, Pistelli A, Cristaudo A. A Novel Panel of Serum Biomarkers for MPM Diagnosis. *Dis Markers* 2017: 6.
41. Johnen G, Gawrych K, Raiko I, Casjens S, Pesch B, Weber DG, Taeger D, Lehnert M, Kollmeier J, Bauer T, Musk AW, Robinson BWS, Bruning T, Creaney J. Calretinin as a blood-based biomarker for mesothelioma. *Bmc Cancer* 2017; 17: 12.
42. Creaney J, van Bruggen I, Hof M, Segal A, Musk AW, de Klerk N, Horick N, Skates SJ, Robinson BWS. Combined CA125 and mesothelin levels for the diagnosis of malignant mesothelioma. *Chest* 2007; 132: 1239-1246.
43. Muley T, Dienemann H, Herth FJF, Thomas M, Meister M, Schneider J. Combination of Mesothelin and CEA Significantly Improves the Differentiation between Malignant Pleural Mesothelioma, Benign Asbestos Disease, and Lung Cancer. *J Thorac Oncol* 2013; 8: 947-951.
44. Cristaudo A, Bonotti A, Simonini S, Vivaldi A, Guglielmi G, Ambrosino N, Chella A, Lucchi M, Mussi A, Foddis R. Combined Serum Mesothelin and Plasma Osteopontin Measurements in Malignant Pleural Mesothelioma. *J Thorac Oncol* 2011; 6: 1587-1593.
45. Creaney J, Yeoman D, Musk AW, de Klerk N, Skates SJ, Robinson BW. Plasma versus serum levels of osteopontin and mesothelin in patients with malignant mesothelioma--which is best? *Lung Cancer* 2011; 74: 55-60.

46. Creaney J, Sneddon S, Dick IM, Dare H, Boudville N, Musk AW, Skates SJ, Robinson BW. Comparison of the diagnostic accuracy of the MSLN gene products, mesothelin and megakaryocyte potentiating factor, as biomarkers for mesothelioma in pleural effusions and serum. *Dis Markers* 2013; 35: 119-127.
47. Hollevoet K, Nackaerts K, Thas O, Thimpont J, Gennonpre P, De Vuyst P, Bosquee L, Legrand C, Kellen E, Kishi Y, Delanghe JR, van Meerbeeck JP. The Effect of Clinical Covariates on the Diagnostic and Prognostic Value of Soluble Mesothelin and Megakaryocyte Potentiating Factor. *Chest* 2012; 141: 477-484.
48. Hooper CE, Morley AJ, Virgo P, Harvey JE, Kahan B, Maskell NA. A prospective trial evaluating the role of mesothelin in undiagnosed pleural effusions. *Eur Resp J* 2013; 41: 18-24.
49. Corradi M, Goldoni M, Alinovi R, Tiseo M, Ampollini L, Bonini S, Carbognani P, Casalini A, Mutti A. YKL-40 and Mesothelin in the Blood of Patients with Malignant Mesothelioma, Lung Cancer and Asbestosis. *Anticancer Res* 2013; 33: 5517-5524.
50. Demir M, Kaya H, Taylan M, Ekinci A, Yilmaz S, Teke F, Sezgi C, Tanrikulu AC, Meteroglu F, Senyigit A. Evaluation of New Biomarkers in the Prediction of Malignant Mesothelioma in Subjects with Environmental Asbestos Exposure. *Lung* 2016; 194: 409-417.
51. Maeda R, Tabata C, Tabata R, Eguchi R, Fujimori Y, Nakano T. Is Serum Thioredoxin-1 a Useful Clinical Marker for Malignant Pleural Mesothelioma? *Antioxid Redox Signal* 2011; 15: 685-689.
52. Holdenrieder S, Hatz R, Reinmiedl J, Hofmann K, Schalhorn A, Stieber P. Improved diagnosis of mesothelioma by a combination of soluble mesothelin-related peptide and CYFRA 21-1. *LaboratoriumsMedizin* 2015; 39: 103-113.
53. Yasumitsu A, Tabata C, Tabata R, Hirayama N, Murakami A, Yamada S, Terada T, Iida S, Tamura K, Fukuoka K, Kuribayashi K, Nakano T. Clinical Significance of Serum Vascular Endothelial Growth Factor in Malignant Pleural Mesothelioma. *J Thorac Oncol* 2010; 5: 479-483.
54. Dongel I, Akbas A, Benli I, Bayram M. Comparison of serum biochemical markers in patients with mesothelioma and pleural plaques versus healthy individuals exposed to environmental asbestos. *Turk Gogus Kalp Damar Cerrahisi Derg* 2019; 27: 374-380.
55. Watzka SB, Posch F, Pass HI, Huflejt M, Bernhard D, Hannigan GE, Muller MR. Detection of integrin-linked kinase in the serum of patients with malignant pleural mesothelioma. *J Thorac Cardiovasc Surg* 2011; 142: 384-389.
56. Watzka SB, Posch F, Pass HI, Flores RM, Hannigan GE, Bernhard D, Weber M, Mueller MR. Serum concentration of integrin-linked kinase in malignant pleural mesothelioma and after asbestos exposure. *Eur J Cardio-Thorac Surg* 2013; 43: 940-944.
57. Alatas F, Alatas O, Metintas M, Colak O, Harmanci E, Demir S. Diagnostic value of CEA, CA 15-3, CA 19-9, CYFRA 21-1, NSE and TSA assay in pleural effusions. *Lung Cancer* 2001; 31: 9-16.
58. Fuhrman C, Duche JC, Chouaid C, Abd Alsamad I, Atassi K, Monnet I, Tillement JP, Housset B. Use of tumor markers for differential diagnosis of mesothelioma and secondary pleural malignancies. *Clin Biochem* 2000; 33: 405-410.
59. Ostroff RM, Mehan MR, Stewart A, Ayers D, Brody EN, Williams SA, Levin S, Black B, Harbut M, Carbone M, Goparaju C, Pass HI. Early Detection of Malignant Pleural Mesothelioma in Asbestos-Exposed Individuals with a Noninvasive Proteomics-Based Surveillance Tool. *Plos One* 2012; 7: 10.
60. Schneider MA, Muley T, Kahn NC, Warth A, Thomas M, Herth FJF, Dienemann H, Meister M. Glycodelin is a potential novel follow-up biomarker for malignant pleural mesothelioma. *Oncotarget* 2016; 7: 71285-71297.

61. Zhang XF, Shen WK, Dong XM, Fan JP, Liu LX, Gao X, Kernstine KH, Zhong L. Identification of Novel Autoantibodies for Detection of Malignant Mesothelioma. *Plos One* 2013; 8: 9.
62. Creaney J, Dick IM, Musk AW, Olsen NJ, Robinson BWS. Immune response profiling of malignant pleural mesothelioma for diagnostic and prognostic biomarkers. *Biomarkers* 2016; 21: 551-561.
63. Cerciello F, Choi M, Nicastri A, Bausch-Fluck D, Ziegler A, Vitek O, Felley-Bosco E, Stahel R, Aebersold R, Wollscheid B. Identification of a seven glycopeptide signature for malignant pleural mesothelioma in human serum by selected reaction monitoring. *Clin Proteom* 2013; 10: 12.
64. Santarelli L, Strafella E, Staffolani S, Amati M, Emanuelli M, Sartini D, Pozzi V, Carbonari D, Bracci M, Pignotti E, Mazzanti P, Sabbatini A, Ranaldi R, Gasparini S, Neuzil J, Tomasetti M. Association of MiR-126 with Soluble Mesothelin-Related Peptides, a Marker for Malignant Mesothelioma. *Plos One* 2011; 6: 9.
65. Bononi I, Comar M, Puozzo A, Stendardo M, Boschetto P, Orecchia S, Libener R, Guaschino R, Pietrobon S, Ferracin M, Negrini M, Martini F, Bovenzi M, Tognon M. Circulating microRNAs found dysregulated in ex-exposed asbestos workers and pleural mesothelioma patients as potential new biomarkers. *Oncotarget* 2016; 7: 82700-82711.
66. Matboli M, Shafei AE, Azazy AEM, Reda M, El-Khazragy N, Nagy AA, Ali MA, Sobhi M, Abdel-Rahman O. Clinical evaluation of circulating miR-548a-3p and-20a expression in malignant pleural mesothelioma patients. *Biomark Med* 2018; 12: 129-139.
67. Tomasetti M, Staffolani S, Nocchi L, Neuzil J, Strafella E, Manzella N, Mariotti L, Bracci M, Matteo VA, Amati M, Santarelli L. Clinical significance of circulating miR-126 quantification in malignant mesothelioma patients. *Clin Biochem* 2012; 45: 575-581.
68. Matboli M, Shafei AE, Ali MA, Gaber AI, Galal A, Tarek O, Marei M, Khairy E, El-Khazragy N, Anber N, Abdel-Rahman O. Clinical significance of serum DRAM1 mRNA, ARSA mRNA, hsa-miR-2053 and lncRNA-RP1-86D1.3 axis expression in malignant pleural mesothelioma. *J Cell Biochem* 2019; 120: 3203-3211.
69. Muraoka T, Soh J, Toyooka S, Aoe K, Fujimoto N, Hashida S, Maki Y, Tanaka N, Shien K, Furukawa M, Yamamoto H, Asano H, Tsukuda K, Kishimoto T, Otsuki T, Miyoshi S. The degree of microRNA-34b/c methylation in serum-circulating DNA is associated with malignant pleural mesothelioma. *Lung Cancer* 2013; 82: 485-490.
70. Sato H, Soh J, Aoe K, Fujimoto N, Tanaka S, Namba K, Torigoe H, Shien K, Yamamoto H, Tomida S, Tao H, Okabe K, Kishimoto T, Toyooka S. Droplet digital PCR as a novel system for the detection of microRNA-34b/c methylation in circulating DNA in malignant pleural mesothelioma. *Int J Oncol* 2019; 54: 2139-2148.
71. Aguilar-Madrid G, Pesch B, Calderon-Aranda ES, Burek K, Jimenez-Ramirez C, Juarez-Perez CA, Ochoa-Vazquez MD, Torre-Bouscoulet L, Acosta-Saavedra LC, Sada-Ovalle I, Garcia-Figueroa J, Alvarado-Cabrero I, Castillo-Gonzalez P, Baez-Saldana AR, Perez-Padilla JR, Osnaya-Juarez J, Rivera-Rosales RM, Garcia-Bazan EM, Bautista-Aragon YL, Lazcano-Hernandez E, Munguia-Canales DA, Argote-Greene LM, Taeger D, Weber DG, Casjens S, Raiko I, Bruning T, Johnen G. Biomarkers for Predicting Malignant Pleural Mesothelioma in a Mexican Population. *Int J Med Sci* 2018; 15: 883-891.
72. Jimenez-Ramirez C, Casjens S, Juarez-Perez CA, Raiko I, Del Razo LM, Taeger D, Calderon-Aranda ES, Rihs HP, Acosta-Saavedra LC, Weber DG, Cabello-Lopez A, Pesch B, Ochoa-Vazquez MD, Burek K, Torre-Bouscoulet L, Perez-Padilla JR, Garcia-Bazan EM, Bruning T, Johnen G, Aguilar-Madrid G. Mesothelin, Calretinin, and Megakaryocyte Potentiating Factor as Biomarkers of Malignant Pleural Mesothelioma. *Lung* 2019; 197: 641-649.

73. Johnen G, Burek K, Raiko I, Wichert K, Pesch B, Weber DG, Lehnert M, Casjens S, Hagemeyer O, Taeger D, Bruning T, MoMar Study G. Prediagnostic detection of mesothelioma by circulating calretinin and mesothelin - a case-control comparison nested into a prospective cohort of asbestos-exposed workers. *Scientific Reports* 2018; 8: 11.
74. Creaney J, Dick IM, Meniawy TM, Leong SL, Leon JS, Demelker Y, Segal A, Musk AW, Lee YCG, Skates SJ, Nowak AK, Robinson BWS. Comparison of fibulin-3 and mesothelin as markers in malignant mesothelioma. *Thorax* 2014; 69: 895-902.
75. Pass HI, Levin SM, Harbut MR, Melamed J, Chiriboga L, Donington J, Huflejt M, Carbone M, Chia D, Goodglick L, Goodman GE, Thornquist MD, Liu G, de Perrot M, Tsao MS, Goparaju C. Fibulin-3 as a blood and effusion biomarker for pleural mesothelioma. *N Engl J Med* 2012; 367: 1417-1427.
76. Jiang ZQ, Ying SB, Shen W, He XL, Chen JQ, Xia HL, Yu M, Xiao Y, Feng LF, Zhu LJ, Ju L, Guo XN, Zhang YX, Shen JW, Tong Y, Zhang X, Lou JL. Plasma Fibulin-3 as a Potential Biomarker for Patients with Asbestos-Related Diseases in the Han Population. *Dis Markers* 2017: 8.
77. Paleari L, Rotolo N, Imperatori A, Puzone R, Sessa F, Franzi F, Meacci E, Camplese P, Cesario A, Paganuzzi M. Osteopontin is not a specific marker in malignant pleural mesothelioma. *Int J Biol Markers* 2009; 24: 112-117.
78. Amati M, Tomasetti M, Scartozzi M, Mariotti L, Alleva R, Pignotti E, Borghi B, Valentino M, Governa M, Neuzil J, Santarelli L. Profiling tumor-associated markers for early detection of malignant mesothelioma: an epidemiologic study. *Cancer Epidemiol Biomarkers Prev* 2008; 17: 163-170.
79. Weber DG, Gawrych K, Casjens S, Brik A, Lehnert M, Taeger D, Pesch B, Kollmeier J, Bauer TT, Johnen G, Brning T. Circulating miR-132-3p as a Candidate Diagnostic Biomarker for Malignant Mesothelioma. *Dis Markers* 2017: 15.
80. Weber DG, Johnen G, Bryk O, Jockel KH, Bruning T. Identification of miRNA-103 in the Cellular Fraction of Human Peripheral Blood as a Potential Biomarker for Malignant Mesothelioma - A Pilot Study. *Plos One* 2012; 7: 9.
81. Weber DG, Casjens S, Johnen G, Bryk O, Raiko I, Pesch B, Kollmeier J, Bauer TT, Bruning T. Combination of MiR-103a-3p and Mesothelin Improves the Biomarker Performance of Malignant Mesothelioma Diagnosis. *Plos One* 2014; 9: 15.
82. Smeele P, d'Almeida SM, Meiller C, Chene AL, Liddell C, Cellerin L, Montagne F, Deshayes S, Benziane S, Copin MC, Hofman P, Le Pimpec-Barthes F, Porte H, Scherpereel A, Gregoire M, Jean D, Blanquart C. Brain-derived neurotrophic factor, a new soluble biomarker for malignant pleural mesothelioma involved in angiogenesis. *Mol Cancer* 2018; 17: 148.
83. Davies HE, Sadler RS, Bielsa S, Maskell NA, Rahman NM, Davies RJO, Ferry BL, Lee YCG. Clinical Impact and Reliability of Pleural Fluid Mesothelin in Undiagnosed Pleural Effusions. *Am J Respir Crit Care Med* 2009; 180: 437-444.
84. Creaney J, Segal A, Olsen N, Dick IM, Musk AW, Skates SJ, Robinson BW. Pleural fluid mesothelin as an adjunct to the diagnosis of pleural malignant mesothelioma. *Dis Markers* 2014; 2014: 413946.
85. Creaney J, Yeoman D, Naumoff LK, Hof M, Segal A, Musk AW, De Klerk N, Horick N, Skates SJ, Robinson BW. Soluble mesothelin in effusions: a useful tool for the diagnosis of malignant mesothelioma. *Thorax* 2007; 62: 569-576.
86. Canessa PA, Ferro P, Manta C, Sivori M, Franceschini MC, Fedeli F, Roncella S. Clinical value of mesothelin in pleural effusions versus histology by medical thoracoscopy: brief report. *Med Oncol* 2013; 30: 649.
87. Canessa PA, Franceschini MC, Ferro P, Battolla E, Dessanti P, Manta C, Sivori M, Pezzi R, Fontana V, Fedeli F, Pistillo MP, Roncella S. Evaluation of soluble mesothelin-related peptide as a diagnostic marker of malignant pleural mesothelioma effusions: its contribution to cytology. *Cancer Invest* 2013; 31: 43-50.

88. Aleman C, Porcel JM, Segura RM, Alegre J, Esquerda A, Ruiz E, Bielsa S, de Sevilla TF. Pleural fluid mesothelin for the differential diagnosis of exudative pleural effusions. *Med Clin* 2009; 133: 449-453.
89. Fujimoto N, Gemba K, Asano M, Wada S, Ono K, Ozaki S, Kishimoto T. Soluble mesothelin-related protein in pleural effusion from patients with malignant pleural mesothelioma. *Exp Ther Med* 2010; 1: 313-317.
90. Mundt F, Nilsonne G, Arslan S, Csuros K, Hillerdal G, Yildirim H, Metintas M, Dobra K, Hjerpe A. Hyaluronan and N-ERC/mesothelin as key biomarkers in a specific two-step model to predict pleural malignant mesothelioma. *PLoS One* 2013; 8: e72030.
91. Battolla E, Canessa PA, Ferro P, Franceschini C, Fontana V, Dessanti P, Pinelli V, Morabito A, Fedeli F, Pistillo MP, Roncella S. Comparison of the Diagnostic Performance of Fibulin-3 and Mesothelin in Patients with Pleural Effusions from Malignant Mesothelioma. *Anticancer Res* 2017; 37: 1387-1391.
92. Blanquart C, Gueugnon F, Nguyen JM, Roulois D, Cellerin L, Sagan C, Perigaud C, Scherpereel A, Gregoire M. CCL2, galectin-3, and SMRP combination improves the diagnosis of mesothelioma in pleural effusions. *J Thorac Oncol* 2012; 7: 883-889.
93. Wang XF, Wu YH, Wang MS, Wang YS. CEA, AFP, CA125, CA153 and CA199 in malignant pleural effusions predict the cause. *Asian Pac J Cancer Prev* 2014; 15: 363-368.
94. Suzuki H, Hirashima T, Kobayashi M, Sasada S, Okamoto N, Uehara N, Tamiya M, Matsuura Y, Morishita N, Kawase I. Cytokeratin 19 fragment/carcinoembryonic antigen ratio in pleural effusion is a useful marker for detecting malignant pleural mesothelioma. *Anticancer Res* 2010; 30: 4343-4346.
95. Paganuzzi M, Onetto M, Marroni P, Filiberti R, Tassara E, Parodi S, Felletti R. Diagnostic value of CYFRA 21-1 tumor marker and CEA in pleural effusion due to mesothelioma. *Chest* 2001; 119: 1138-1142.
96. Filiberti R, Parodi S, Libener R, Ivaldi GP, Canessa PA, Ugolini D, Bobbio B, Marroni P. Diagnostic value of mesothelin in pleural fluids: comparison with CYFRA 21-1 and CEA. *Med Oncol* 2013; 30: 543.
97. Otoshi T, Kataoka Y, Ikegaki S, Saito E, Matsumoto H, Kaku S, Shimada M, Hirabayashi M. Pleural effusion biomarkers and computed tomography findings in diagnosing malignant pleural mesothelioma: A retrospective study in a single center. *PLoS One* 2017; 12: e0185850.
98. Porcel JM, Esquerda A, Martinez-Alonso M, Bielsa S, Salud A. Identifying Thoracic Malignancies Through Pleural Fluid Biomarkers A Predictive Multivariate Model. *Medicine* 2016; 95: 8.
99. Hegmans J, Veltman JD, Fung ET, Verch T, Glover C, Zhang FJ, Allard WJ, T'Jampens D, Hoogsteden HC, Lambrecht BN, Aerts J. Protein Profiling of Pleural Effusions to Identify Malignant Pleural Mesothelioma Using SELDI-TOF MS. *Technol Cancer Res Treat* 2009; 8: 323-332.
100. Mundt F, Johansson HJ, Forshed J, Arslan S, Metintas M, Dobra K, Lehtio J, Hjerpe A. Proteome screening of pleural effusions identifies galectin 1 as a diagnostic biomarker and highlights several prognostic biomarkers for malignant mesothelioma. *Mol Cell Proteomics* 2014; 13: 701-715.
101. Porcel JM, Esquerda A, Rodriguez-Panadero F, Martinez-Iribarren A, Bielsa S. The use of pleural fluid sCD44v6/std ratio for distinguishing mesothelioma from other pleural malignancies. *J Thorac Oncol* 2011; 6: 190-194.